

Modality-specific reactive and predictive responses in the human superior colliculus

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Danlei Chen^{1,2}✉, Maya Barton-Zuckerman³, Zoe Kross³, Philip A. Kragel⁴, Paul W. Savoca⁵, Lawrence L. Wald², Marta Bianciardi^{2,6}, Tor D. Wager⁷, Ji-Kyung Choi⁸, Jiahe Zhang⁹, Karen S. Quigley^{3,9}, Ajay B. Satpute^{2,5,11}, Lisa Feldman Barrett^{2,3,9,11}✉ & Jordan E. Theriault-Brown^{2,3,10,11}✉

Inspired by recent theoretical work emphasizing predictive neural processing, we tested whether a midbrain structure for multimodal sensory integration—the superior colliculus—would show predictive responses to both visual and somatosensory stimulation. A multi-stage experimental design allowed us to test predictive activity in the human superior colliculus while 7-Tesla functional magnetic resonance imaging (fMRI) allowed us to map its sensory response across superior colliculus layers ($N = 80$). As expected from non-human animal work, visual and somatosensory stimulation elicited distinct depth-dependent responses. We also observed a predictive response: these same modality-specific and depth-dependent effects occurred during task periods preceding direct stimulation, when sensorimotor input was identical between visual and somatosensory conditions. Further analyses rule out cross-trial carryover and show corresponding effects in the thalamic sensory nuclei. Here our results confirm that, in humans, the superior colliculus is sensitive to both visual and somatosensory input, and they demonstrate that it carries predictive signals for upcoming sensory events.

The superior colliculus, as the midbrain target of the optic nerve, has long been recognized as a crucial midbrain structure for visual attention and visually guided behaviors¹. Non-human animal evidence also suggests that the superior colliculus plays an important role in non-visual (for example, auditory and somatosensory) processing. In non-human vertebrates, single-unit recordings demonstrate visual response among neurons in the superficial layer of the superior colliculus (organized into retinotopic receptive fields), yet neurons in intermediate and deep layers are sensitive to visual, auditory and somatosensory stimulation

and to combinations of these inputs^{2–5}. Furthermore, neurons in intermediate and deep layers of the superior colliculus play a role in controlling motor movements of the eyes, head and body^{1,6,7}. The consensus perspective from non-human vertebrate research is that the superior colliculus serves multiple functions in sensory processing, integration and control, which are organized at a fine scale across its anatomical layers. Although some evidence for similar functional organization exists in humans^{8–12}, comprehensive validation remains limited, in large part because superior colliculus layers are difficult to spatially resolve

¹Research Laboratory of Electronics, Massachusetts Institute of Technology, Cambridge, MA, USA. ²Department of Radiology, Athinoula A. Martinos Center for Biomedical Imaging, Massachusetts General Hospital, Boston, MA, USA. ³Department of Psychology, Northeastern University, Boston, MA, USA. ⁴Department of Psychology, Emory University, Atlanta, GA, USA. ⁵Department of Psychology, University of California, Los Angeles, Los Angeles, CA, USA. ⁶Division of Sleep Medicine, Harvard University, Boston, MA, USA. ⁷Department of Psychological and Brain Sciences, Dartmouth College, Hanover, NH, USA. ⁸Department of Surgery, University of California, San Francisco, San Francisco, CA, USA. ⁹Department of Psychiatry, Massachusetts General Hospital, Boston, MA, USA. ¹⁰Department of Biology, Northeastern University, Boston, MA, USA. ¹¹These authors jointly supervised this work: Ajay B. Satpute, Lisa Feldman Barrett, Jordan E. Theriault-Brown. ✉e-mail: danlei@mit.edu; l.barrett@northeastern.edu; jo.theriault@northeastern.edu

using standard resolutions in 3-Tesla fMRI. More importantly, however, this consensus perspective has largely focused on how the superior colliculus reacts to sensory stimuli, whereas emerging integrative perspectives on whole-brain function (that is, allostatic frameworks^{13–17} and predictive processing frameworks^{18–20}) suggest that the brain predicts and prepares for sensory input that has not yet occurred. The present study both validates the consensus perspective in humans and extends it, using high-resolution 7-Tesla fMRI to show that the superior colliculus blood-oxygen-level-dependent (BOLD) response is organized across layers, sensitive to both visual and somatosensory stimulation, and occurs both in response to and in anticipation of visual and somatosensory stimulation.

An allostatic perspective helps contextualize why sensory integration was previously observed in the superior colliculus^{21–23} (for review, see ref. 7) and makes further predictions about its functions. Allostasis refers to predictive (as opposed to error-based) regulation, which allows the brain to leverage learned experience to avoid costly mistakes^{16,17}. On this account, and in related predictive processing accounts^{13,18–20}, the brain, as a whole, predicts sensory input using generalizations informed by past experience (that is, an internal model)—anticipating the sensory consequences (for example, visual and somatosensory) of a motor command^{13,20,24}. If this is correct, then it follows that the superior colliculus, and other brain regions involved in motor control, should be sensitive to sensory predictions. Some initial evidence supports this hypothesis. For example, superior colliculus activity is observed as motor commands reduce the discrepancy between expected and received sensory input (that is, prediction error) by adjusting saccades closer to the predicted target location²⁵. Furthermore, some visual-sensitive neurons in the deep layers of the superior colliculus are also active during the preparation and execution of motor movements^{26,27}. However, superior colliculus predictive activity has largely been within the purview of motor planning research, and sensory processing work has focused on superior colliculus reactivity. By contrast, in the present work, we show that the superior colliculus BOLD response is sensitive to anticipated sensory stimulation along established anatomical gradients, suggesting that the superior colliculus participates in distributed predictive networks that represent expected sensory input before it arrives at sensory surfaces.

To examine the fine-scale layer structures of the human superior colliculus and its role in sensory prediction, we analyzed a 7-Tesla fMRI dataset ($N = 80$) that was originally collected by our group to test subcortical contributions to reinforcement learning (Methods). A simple task was repeated in five runs (120 trials in total) and involved neutral or aversive visual or somatosensory stimulation (between groups; $n = 40$ each; Fig. 1a). Visual stimulation used normed affective images (International Affective Picture System (IAPS)²⁸), and somatosensory

stimulation used mechanical pressure to the left thumb (Fig. 1b). Stimulus aversiveness was under partial control by subjects: before the stimulus was presented, subjects chose between two shapes (or, in 50% of trials, the computer chose for them), each with some probability of triggering aversive feedback. The probability of aversive feedback was 50% for each shape on average but changed in a random walk across trials (and is not explored in the present study). Shapes were presented ('shape presentation'), after which a double-sided arrow appeared, signaling that subjects should make a decision with the button box using their right hand. Computer-based decisions were indicated by the appearance of a one-sided arrow. The period from the appearance of the double-sided or single-sided arrow was distinguished from the rest of the shape presentation period as the 'pre-stimulus decision' period. This pre-stimulus decision period was modeled to test whether the BOLD response in the superior colliculus was sensitive to the anticipated sensory stimulation. Thus, we analyzed superior colliculus response in three periods: shape presentation, pre-stimulus decision and sensory stimulation. The high spatial resolution afforded by 7-Tesla imaging also allowed us to map the layer structure within the superior colliculus more accurately, providing a greater level of spatial specificity compared to prior 3-Tesla work examining superior colliculus function in humans^{8–12}. The superior colliculus is approximately 6 mm in diameter in each cerebral hemisphere, and neighboring regions include the periaqueductal gray (PAG) and cerebrospinal fluid (CSF), both of which must be carefully parsed from the region of interest (ROI)^{29–31}. Our voxel-wise resolution was 1.1 mm isotropic (versus 2–3 mm, which is standard at 3 Tesla), which allowed us to examine average differences across the superior colliculus' superficial and deep layers.

Results

Preprocessing and alignment of the superior colliculus across subjects

After preprocessing (Methods), subject-level contrast maps were estimated for shape presentation, pre-stimulus decision and sensory stimulation periods (Fig. 1a). Montreal Neurological Institute (MNI) normalization is not optimized for small subcortical structures such as the superior colliculus, so, to improve superior colliculus alignment across subjects, we combined predefined ROI boundaries with subject-specific PAG and CSF masks to create subject-specific superior colliculus masks (Methods). These subject-specific superior colliculus masks were aligned at the group level using DARTEL³², which substantially improved subject overlap (Methods). Transformations were applied to the original contrast maps to align all functional data to a common space, optimized for overlap of the superior colliculus ROIs. For voxel-wise volumetric analyses within the superior colliculus ROI (for example, Fig. 1d), 3-mm full width at half maximum (FWHM) smoothing

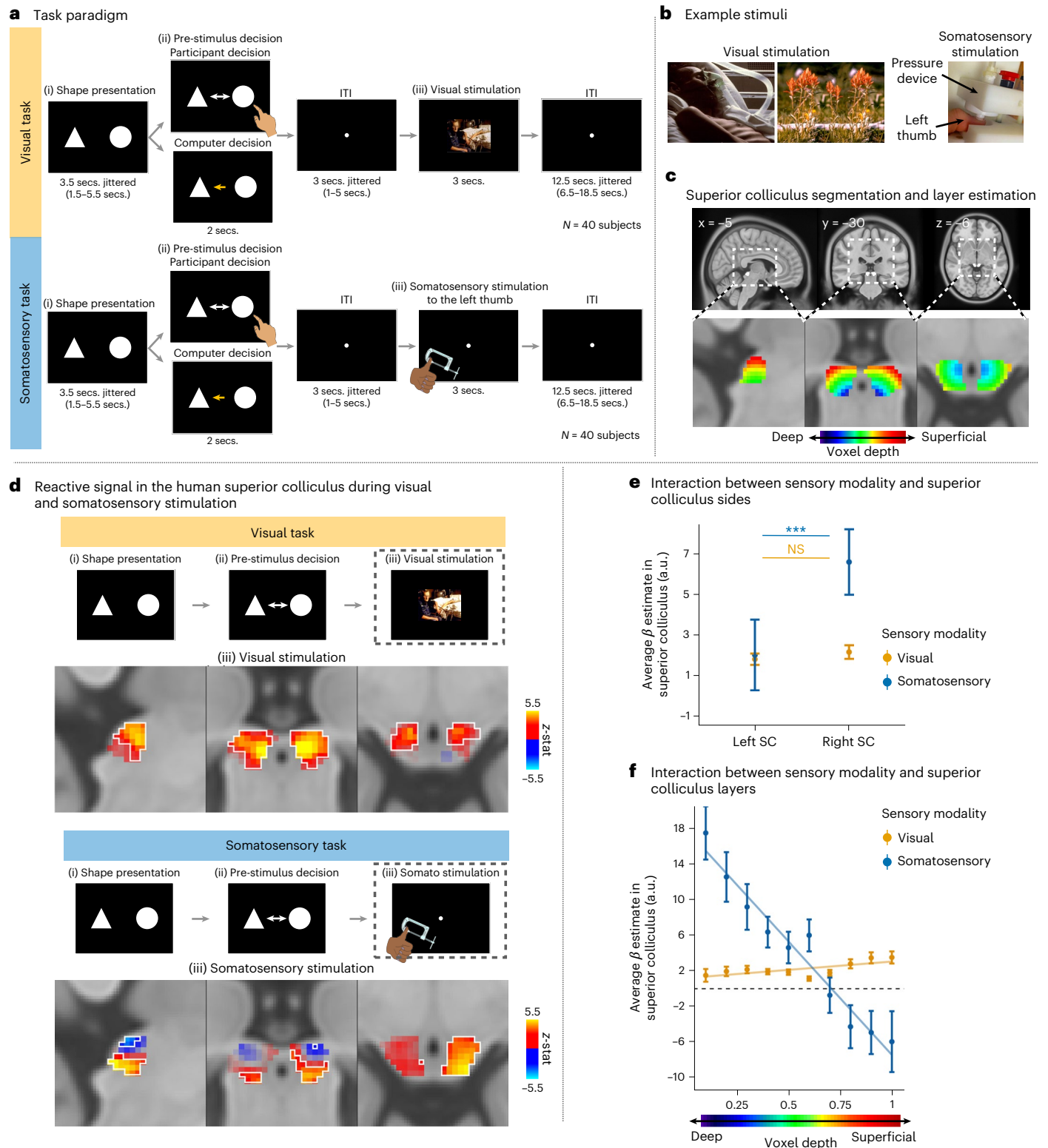
Fig. 1 | Task paradigm and reactive signal in the superior colliculus during sensory stimulation. a, Task paradigm. The experiment was originally designed as a probabilistic avoidance learning task. In the shape presentation period, subjects viewed two visual cues. In the pre-stimulus decision period, either subjects were asked to choose one of the shapes (that is, participant decision; 50% of trials) or a shape was randomly chosen for them (that is, computer decision; Methods). After a jittered ITI, in the visual task, an image was displayed; in the somatosensory task, mechanical pressure was delivered to the left thumb. Trials were separated by a jittered ITI. **b**, Example stimuli for visual stimulation and experimental apparatus for somatosensory stimulation. During visual stimulation, subjects viewed images. During somatosensory stimulation, mechanical pressure was applied to the nail bed on the left thumb. **c**, Group-level superior colliculus mask. Voxel depth was estimated for the layer structure within the superior colliculus from deep to superficial. **d**, Superior colliculus BOLD response during visual and somatosensory stimulation. Colored voxels refer to z -statistics (task period > baseline). White outlines are superimposed onto clusters that passed a voxel-wise cutoff of $q < 0.05$ (one-tailed, FDR corrected) within the group-level superior colliculus mask. Additional slices of the superior colliculus maps are shown in Extended Data Fig. 3. **e**, During

somatosensory stimulation, the right superior colliculus showed greater BOLD response than the left, consistent with mechanical pressure applied contralaterally on the left thumb. By contrast, visual stimulation did not show a significant difference in BOLD response between the left and right superior colliculus (side $\times$ sensory modality interaction from linear mixed-effect model, $F_{1,67,29} = 9.37, P = 0.003$). **f**, The superior colliculus showed greater BOLD response in the superficial layers compared to the deep layers during visual stimulation and greater BOLD response in the deep layers compared to superficial layers during somatosensory stimulation (voxel depth $\times$ sensory modality interaction from linear mixed-effect model, $F_{1,65,79} = 30.17, P < 0.001$). Note that the y axis range in **e** and **f** is set to accommodate the larger range of BOLD response for somatosensory stimulation, which visually compresses the main effect across voxel depth for visual stimulation. Linear regression lines were fitted to the average BOLD response across voxel depth for visual and somatosensory tasks, separately. Error bars indicate ± 1 s.e.m. across subjects. Two-sided paired t -tests: *** $P < 0.001$; NS, not significant. For all panels, $n = 40$ subjects per modality group (visual and somatosensory; 80 total). SC, superior colliculus; secs., seconds; Somato, somatosensory.

was applied within the boundaries of the group-level ROI mask. For linear mixed-effects model (for example, Extended Data Table 1), average β estimates (for example, Figs. 1e,f and 2) and multivariate classification analyses (Fig. 3), no smoothing was applied. For group-level whole-brain analyses, 3-mm FWHM smoothing was applied. Whole-brain BOLD response at the group level showed expected patterns (for example, visual cortex during visual stimulation and right primary somatosensory cortex during somatosensory stimulation of the left thumb; Extended Data Fig. 1).

Visual and somatosensory stimulation elicit spatially distinct patterns of BOLD response across layers of the superior colliculus

Consistent with prior non-human animal work^{2,3,5}, superficial layers of the superior colliculus were sensitive to visual processing, whereas deeper layers appeared to be sensitive to both visual and somatosensory processing. Voxels were labeled according to their depth (Fig. 1c and Methods) in order to examine relative BOLD changes from superficial to deep layers of the superior colliculus. In the sensory stimulation period,



both visual and somatosensory stimulation elicited significant changes within the superior colliculus, relative to baseline fixation (Fig. 1d and Extended Data Figs. 2 and 3). Of note, visual and somatosensory stimulation elicited distinct patterns of BOLD response across voxel depth (that is, superior colliculus layers). Interactions between sensory modality and voxel depth (operationalized as 10 ordinal bins; Methods) were estimated in a linear mixed-effects regression, including fixed effects of voxel depth (within subjects; bins 1–10), lateral side (within subjects; left/right) and sensory modality (between subjects; visual/somatosensory) as well as relevant interactions (Methods; for full model results, see Extended Data Table 1a). Two interactions were significant. The first functioned as a sanity check: A significant lateral side $\times$ sensory modality interaction ($F_{1,67.29} = 9.37$, $P = 0.003$; Fig. 1e) demonstrated that the BOLD response to visual stimulation showed no significant lateral differences in superior colliculus ($M_{\text{visual:left}} = 1.83$, $M_{\text{visual:right}} = 2.19$, two-sided paired t -test: $t_{39} = -1.24$, $P = 0.222$), whereas somatosensory stimulation elicited a greater BOLD response in the right (versus left) superior colliculus ($M_{\text{somatosensory:left}} = 2.04$, $M_{\text{somatosensory:right}} = 6.60$, two-sided paired t -test: $t_{39} = -3.75$, $P < 0.001$), consistent with a contralateral response to the mechanical pressure applied to the left thumb (Methods). The second significant interaction, between voxel depth and sensory modality, confirmed the distinct depth-based responses to visual and somatosensory stimulation ($F_{1,65.79} = 30.17$, $P < 0.001$; Fig. 1f), such that visual stimulation elicited greater BOLD response in superficial layers (versus deep layers) ($M_{\text{visual:deep}} = 1.50$, $M_{\text{visual:superficial}} = 3.51$, two-sided paired t -test: $t_{39} = -2.57$, $P = 0.014$), whereas somatosensory stimulation elicited greater BOLD response in deep layers (versus superficial) ($M_{\text{somatosensory:deep}} = 17.48$, $M_{\text{somatosensory:superficial}} = -5.95$, two-sided paired t -test: $t_{39} = 4.89$, $P < 0.001$).

Anticipated visual and somatosensory stimulation also elicit spatially distinct patterns of BOLD response across layers of the superior colliculus

A similar layer-dependent distinction between visual and somatosensory stimuli was observed in the task period before sensory stimuli were delivered (that is, when the present sensory input was identical between sensory conditions). We took this as evidence of an anticipatory BOLD response in the superior colliculus. Before sensory stimulation, subjects observed a pair of shapes (shape presentation period; 3.5-second duration) and then either chose a shape or had a shape chosen for them by the computer (pre-stimulus decision period; 2-second duration) (Methods and Fig. 1a). Superior colliculus BOLD response in the pre-stimulus decision period was estimated using a linear mixed-effects regression, including fixed effects of voxel depth (within subjects; bins 1–10), lateral side (within subjects; left/right), decision type (within subjects; participant/computer) and sensory modality of stimulation (between subjects; visual/somatosensory) as well as relevant interactions (Methods and Extended Data Figs. 2 and 3; for full model results, see Extended Data Table 1a). We observed a significant interaction between voxel depth and the upcoming sensory

modality ($F_{1,66.76} = 30.88$, $P < 0.001$; Fig. 2a), showing that, before visual stimulation occurred, the superior colliculus BOLD response did not differ significantly between deep and superficial layers ($M_{\text{visual:deep}} = 1.48$, $M_{\text{visual:superficial}} = 4.79$, two-sided paired t -test: $t_{39} = -1.53$, $P = 0.134$), whereas, before somatosensory stimulation occurred, the superior colliculus BOLD response was greater in deep (versus superficial) layers ($M_{\text{somatosensory:deep}} = 2.98$, $M_{\text{somatosensory:superficial}} = -3.68$, two-sided paired t -test: $t_{39} = 6.10$, $P < 0.001$) (for more slice visualization, see Extended Data Fig. 3). We also observed a significant main effect of decision type (Extended Data Fig. 4, Supplementary Note 1 and Supplementary Fig. 1).

It was possible that this apparent anticipatory BOLD response in the superior colliculus was not anticipatory but, rather, was a carryover effect from previous trials (for example, visual stimulation in the last trial or a residual sore thumb from previous thumb presses). To rule this out, we modeled superior colliculus response during the shape presentation period, which occurred between the pre-stimulus decision period and the sensory stimulation period from the previous trial (Fig. 1a). Superior colliculus BOLD response in the shape presentation period was estimated using a linear mixed-effects regression, including fixed effects of voxel depth (within subjects; bins 1–10), lateral side (within subjects; left/right) and sensory modality of stimulation (between subjects; visual/somatosensory) as well as relevant interactions. We did observe a significant interaction between voxel depth and sensory modality ($F_{1,61.68} = 6.92$, $P = 0.011$; Fig. 2b; for full model results, see Extended Data Table 1a); however, the pattern was distinct from what was observed in pre-stimulus decision and sensory stimulation periods (Figs. 1d and 2a). For subjects receiving visual stimulation, the shape presentation period did not elicit a significantly different BOLD response across superior colliculus layers ($M_{\text{visual:deep}} = 2.49$, $M_{\text{visual:superficial}} = 2.65$, two-sided paired t -test: $t_{39} = -0.21$, $P = 0.835$), whereas, for subjects receiving somatosensory stimulation, the shape presentation period elicited a greater BOLD response in superficial (versus deep) superior colliculus layers ($M_{\text{somatosensory:deep}} = 1.12$, $M_{\text{somatosensory:superficial}} = 3.38$, two-sided paired t -test: $t_{39} = -3.03$, $P = 0.004$). This suggests that the pattern of depth-dependent activity observed in pre-stimulus decision and sensory stimulation periods was not consistently present across the entire trial, suggesting that the pattern of results during the pre-stimulus decision period represents an anticipatory response and not a residual response carried over from previous trials.

Validating BOLD response similarity by multivariate pattern analysis

The apparent similarity in spatial BOLD response between ‘reactive’ sensory stimulation and ‘predictive’ pre-stimulus decision periods, reported above, was confirmed in a multivariate pattern analysis (along with the mutual dissimilarity of these two periods with the shape presentation period). Similarity and dissimilarity were quantified using a supervised linear support vector machine (SVM) classifier (leave-one-subject-out cross-validation; Methods), trained to

Fig. 2 | Superior colliculus BOLD signal showed layer-specific signals consistent with the sensory modality of upcoming stimulation during pre-stimulus decision period but not during shape presentation period.

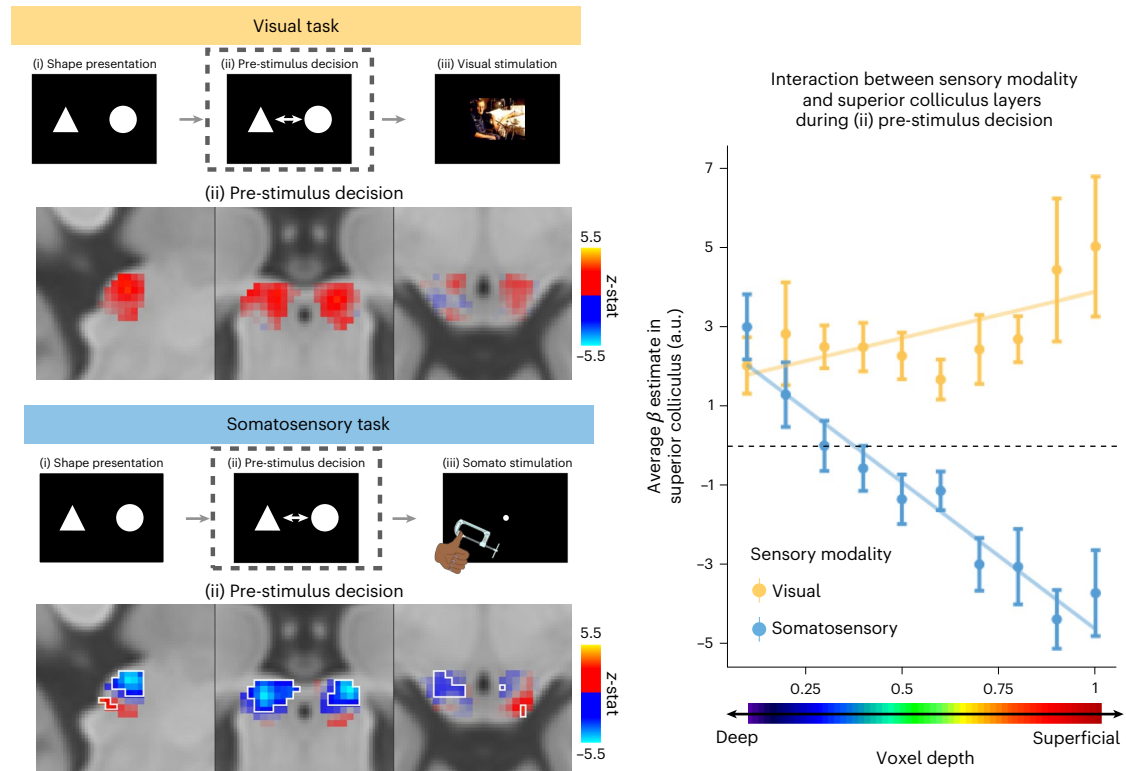
a, During the pre-stimulus decision period, the superior colliculus showed a layer-specific pattern of activity corresponding to the modality of upcoming sensory stimulation (that is, a similar layer-specific pattern to the one observed during sensory stimulation; Fig. 1). Specifically, we observed a modality-by-layer interaction in the superior colliculus: Subjects anticipating somatosensory stimulation showed greater signal in deeper than superficial layers, whereas the visual group showed numerically greater signal in superficial layer but no significant superficial-versus-deep difference (voxel depth $\times$ sensory modality interaction from linear mixed-effect model, $F_{1,66.76} = 30.88$, $P < 0.001$). Note that, although the superior colliculus activity patterns differed between visual and somatosensory groups, the demands during the pre-stimulus decision task period were matched across groups, meaning that only anticipated or

prior task performance could account for the group difference. **b**, Differences in prior task performance were ruled out by analyzing the shape presentation period. In this period, layer-specific BOLD response in superior colliculus was distinct from patterns observed in the pre-stimulus decision or sensory stimulation periods (voxel depth $\times$ sensory modality interaction from linear mixed-effect model, $F_{1,61.68} = 6.92$, $P = 0.011$). Colored voxels refer to z -statistics (task period $>$ baseline). White outlines are superimposed onto clusters that passed a voxel-wise cutoff of $q < 0.05$ (one-tailed, FDR corrected) within the group-level superior colliculus mask. Additional slices of the superior colliculus maps are shown in Extended Data Fig. 3. Linear regression lines were fitted to the average BOLD response across voxel depth for visual and somatosensory tasks, separately. Error bars indicate ± 1 s.e.m. across subjects. For all panels, $n = 40$ subjects per modality group (visual and somatosensory; 80 total). Somato, somatosensory.

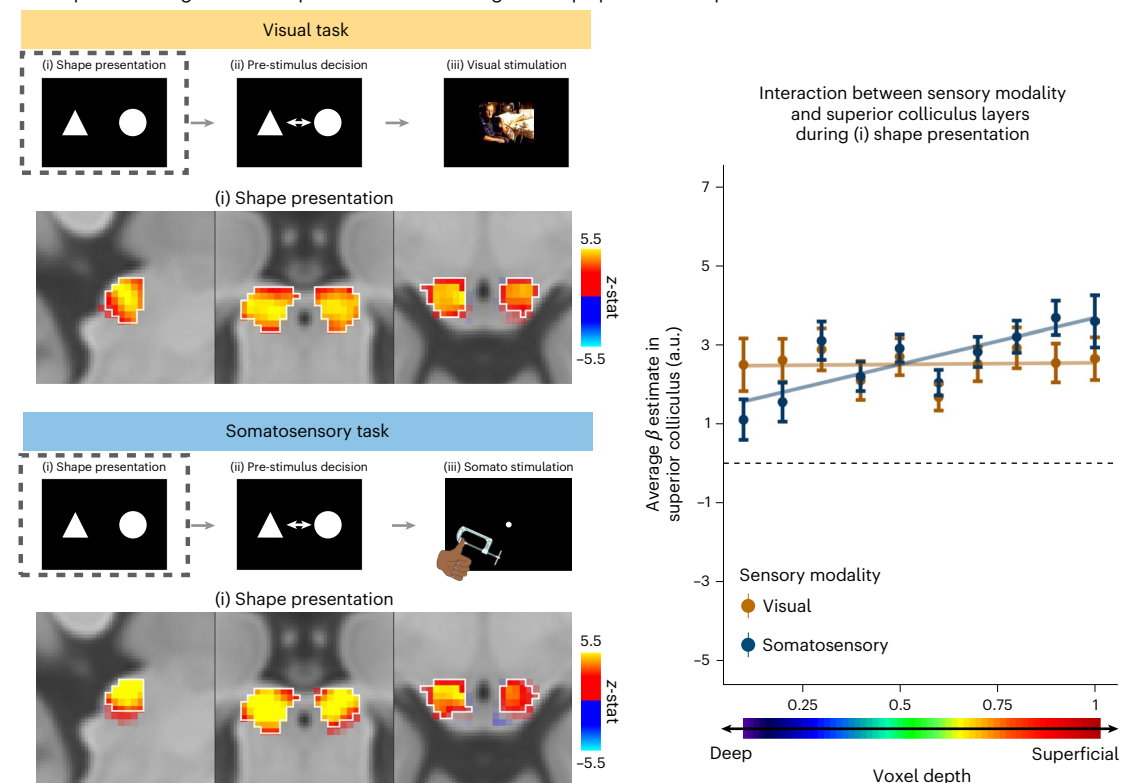
classify contrast images as coming from subjects receiving either visual or somatosensory stimulation. Classifiers were trained on run-level superior colliculus voxel maps (up to five from each subject; Methods) from the pre-stimulus decision period (Fig. 3a), the sensory stimulation period (Fig. 3b) or the shape presentation period

(Fig. 3c) and were tested on run-level maps from a held-out subject. The classifier trained on the pre-stimulus decision period classified held-out pre-stimulus decision period maps significantly above chance (that is, within-category; $M_{\text{accuracy}} = 62.84\%$, s.e.m. = 2.37%, $P < 0.001$, determined using a non-parametric permutation test; Methods) and

a Superior colliculus showed layer-specific pattern of activity predicting upcoming sensory stimulation during pre-stimulus decision period



b No predictive signal in the superior colliculus during the shape presentation period



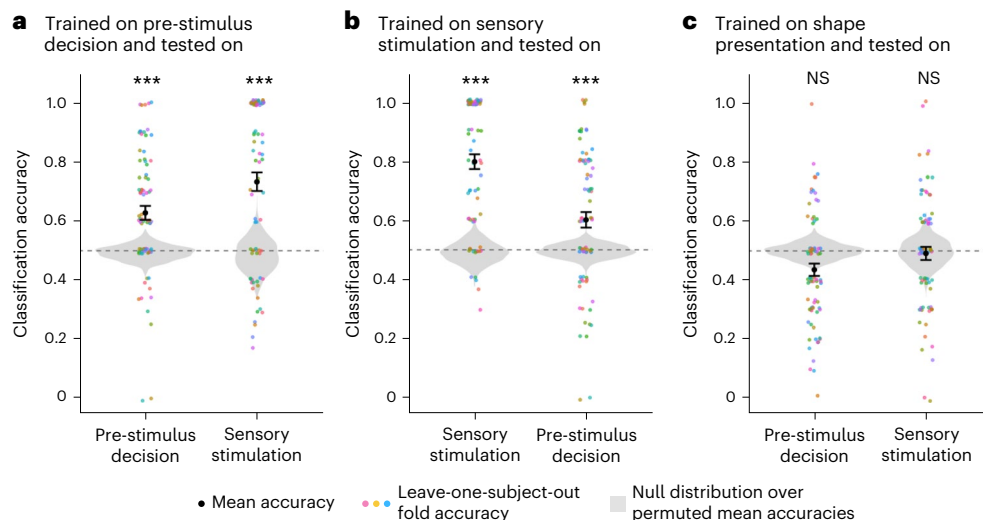


Fig. 3 | Classification accuracy for decoding sensory modality from superior colliculus multi-voxel patterns. **a**, Classifiers trained on run-level superior colliculus maps during pre-stimulus decision period distinguished sensory modality labels above chance in held-out pre-stimulus decision period maps (that is, within-category) and in sensory stimulation period maps (that is, out-of-category), statistically confirming the observed layer-specific similarity of predictive (pre-stimulus decision) and reactive (sensory stimulation) responses within the superior colliculus (Figs. 1 and 2; permutation test, $N = 80$ folds from leave-one-subject-out cross-validation). **b**, Likewise, classifiers trained on sensory stimulation period maps distinguished sensory modality labels above chance in held-out sensory stimulation period maps (that is, within-category) and in pre-stimulus decision period maps (that is, out-of-category; permutation test, $N = 80$ folds from leave-one-subject-out cross-validation). **c**, A different

result was observed for shape presentation maps, where classifiers trained on this period could not distinguish sensory modality labels above chance in either pre-stimulus decision period maps or sensory stimulation period maps, suggesting that the anticipatory BOLD response in the pre-stimulus decision period was not a product of pre-existing BOLD patterns from the prior period (that is, carryover; permutation test, $N = 80$ folds from leave-one-subject-out cross-validation). Each jittered dot represents the mean classification accuracy for one leave-one-subject-out cross-validation fold, averaged across all runs of the held-out subject. Error bars represent the standard error across cross-validation folds. The gray shaded violin shows the null distribution of mean accuracies across all 1,000 permutations, which was used to compute the one-sided P values. The dashed line indicates chance performance (50%). One-sided permutation test: *** $P < 0.001$; NS, not significant.

classified held-out sensory stimulation maps above chance as well (that is, out-of-category; $M_{\text{accuracy}} = 73.35\%$, s.e.m. = 3.10% , $P < 0.001$). Likewise, another classifier, trained on sensory stimulation period maps, classified both sensory stimulation period maps (within-category; $M_{\text{accuracy}} = 79.78\%$, s.e.m. = 2.51% , $P < 0.001$) and pre-stimulus decision period maps (out-of-category) significantly above chance ($M_{\text{accuracy}} = 60.16\%$, s.e.m. = 2.63% , $P < 0.001$).

The third SVM classifier, trained on voxel maps from the shape presentation period, confirmed that the shape presentation period (versus sensory stimulation and pre-stimulus periods) represented a different aggregate spatial pattern of superior colliculus BOLD response. It could not classify out-of-sample visual versus somatosensory subjects in the shape presentation maps (within-category; $M_{\text{accuracy}} = 52.81\%$, s.e.m. = 3.33% , $P = 0.203$) nor could it classify visual versus somatosensory subjects in pre-stimulus decision maps (out-of-category; $M_{\text{accuracy}} = 43.61\%$, s.e.m. = 2.09% , $P = 0.999$) or sensory stimulation maps (out-of-category; $M_{\text{accuracy}} = 49.16\%$, s.e.m. = 2.23% , $P = 0.647$). Given that patterns in the preceding shape presentation period failed to classify visual versus somatosensory subjects, the anticipatory BOLD response observed in the subsequent pre-stimulus decision period is unlikely to simply stem from a residual BOLD pattern carried over from the previous trial.

Predictive and reactive visual and somatosensory responses in thalamic nuclei

If the superior colliculus is sensitive to visual and somatosensory predictive and reactive signals, then an open question is whether more unimodal subcortical regions show a similar pattern of predictive/reactive response. The lateral geniculate nucleus (LGN; Fig. 4a) and ventral posterolateral nucleus (VPL; Fig. 4b) are thalamic nuclei supporting visual and somatosensory processing, respectively: LGN receiving retinal input and interacting with the primary visual cortex and VPL

relaying spinal afferents to the primary somatosensory cortex (see Fig. 27 in ref. 33). We hypothesized that the visual and somatosensory predictive/reactive response in superior colliculus would appear as visual-specific signals in LGN and somatosensory-specific signals in VPL. We ran six linear mixed-effects regressions in total, modeling average BOLD response within the masks of these thalamic nuclei (LGN and VPL; Methods and Fig. 4a,b) and in each task period (sensory stimulation, pre-stimulus decision and shape presentation; Fig. 1a). Each regression included fixed effects of lateral side (within subjects; left/right), decision type (within subjects; participant/computer) and sensory modality of stimulation (between subjects; visual/somatosensory) and all relevant interactions (Methods and Extended Data Table 1b,c).

As expected, LGN was sensitive to both predictive and reactive visual processing. In the sensory stimulation period, the only significant effect was a main effect of sensory modality ($F_{1,74.32} = 54.20$, $P < 0.001$), such that subjects receiving visual stimulation showed a greater BOLD response than subjects receiving somatosensory stimulation (Fig. 4c). Likewise, in the pre-stimulus decision period, the LGN showed only a significant main effect of the modality of upcoming sensory stimulation ($F_{1,65.11} = 36.33$, $P < 0.001$; Fig. 4e), such that subjects about to receive visual stimulation showed greater LGN BOLD response than subjects about to receive somatosensory stimulation. Finally, in the shape presentation period, the BOLD response in LGN did not differ between subjects receiving visual or somatosensory stimulation ($F_{1,78.08} = 0.23$, $P = 0.636$; Fig. 4g).

Also as expected, the VPL was sensitive to both predictive and reactive somatosensory processing. VPL response could be further validated by the lateralization of response, as thumb presses were delivered to the left hand, and should elicit a contralateral VPL response (as also observed in the superior colliculus above; Fig. 1e). In the sensory stimulation period, BOLD in VPL showed a significant interaction between lateral side and sensory modality ($F_{1,1027.31} = 12.19$, $P < 0.001$), such that

BOLD in right VPL exceeded BOLD in left VPL during somatosensory stimulation ($M_{\text{somatosensory: left}} = -5.54$, $M_{\text{somatosensory: right}} = 1.51$, two-sided paired t -test: $t_{39} = -4.75$, $P < 0.001$), whereas, during visual stimulation, no significant lateral difference occurred ($M_{\text{visual: left}} = -1.29$, $M_{\text{visual: right}} = -0.97$, two-sided paired t -test: $t_{39} = -0.93$, $P = 0.358$; Fig. 4d). The VPL pre-stimulus decision period represents a special case, as subjects both deliver a motor response with their right hand (when the shape is not selected by the computer; 50% of trials; Supplementary Note 1 and Extended Data Fig. 4) and may anticipate sensory input to the left thumb (for subjects receiving somatosensory stimulation). Consistent with this special case, VPL BOLD response in the pre-stimulus decision period showed a significant interaction between lateral side and upcoming sensory modality ($F_{1,1137.99} = 7.14$, $P = 0.008$), such that when visual stimulation was anticipated, BOLD response was left-lateralized, contralateral to the motor response ($M_{\text{visual: left}} = 2.40$, $M_{\text{visual: right}} = 0.12$, two-sided paired t -test: $t_{39} = 4.88$, $P < 0.001$; Fig. 4f), whereas, when somatosensory stimulation was anticipated, VPL BOLD response was bilateral ($M_{\text{somatosensory: left}} = 1.88$, $M_{\text{somatosensory: right}} = 1.37$, two-sided paired t -test: $t_{39} = 1.06$, $P = 0.296$; Fig. 4f), consistent with the interpretation that anticipated somatosensory input to the left thumb during somatosensory trials removed the lateralized effect observed during visual trials. Finally, in the shape presentation period, BOLD response in VPL showed no significant main effects or interactions involving the upcoming sensory modality; however, a significant left-lateralized VPL BOLD response was observed ($F_{1,70.27} = 84.80$, $P < 0.001$; Fig. 4h), which is contralateral to the upcoming (right-handed) button press in the shape presentation period and consistent with the interpretation that the VPL predicts and prepares a motor response in the following task period.

Discussion

The superior colliculus has generally been studied in the context of its reaction to sensory stimuli^{2,3,5,12}, yet the present work demonstrated that the superior colliculus (and thalamic nuclei) both react to and anticipate visual and somatosensory stimulation. Furthermore, prior non-human animal work in the superior colliculus identified deep–superficial layer gradients as a critical feature of anatomical organization, and the present work (aided by the spatial resolution of 7-Tesla fMRI) confirmed this gradient in humans and demonstrated that reactive and anticipatory BOLD responses are organized along this biological feature. Specifically, as subjects reacted to visual stimulation (versus subjects receiving somatosensory stimulation), superficial superior colliculus layers showed a greater BOLD response, whereas the opposite pattern (deep > superficial) was observed for somatosensory stimulation (Fig. 1f). As subjects anticipated visual versus somatosensory stimulation, we likewise observed a robust depth-dependent pattern

that differed depending on the anticipated sensory modality (Fig. 2a). These anticipatory visual and somatosensory results occurred in a task period in which task demands (that is, button press and visual input) were identical across groups (pre-stimulus decision period; Fig. 1a), meaning that any difference must be driven by factors preceding or following the task period. Ideally, in-scanner eye tracking would have confirmed whether anticipatory eye movements varied between visual and somatosensory groups (for out-of-scanner eye-tracking experiment, see Supplementary Note 2); however, we still observed differences across sensory groups despite identical sensory and motor inputs, supporting the interpretation that these responses reflect modality-specific predictive processes rather than purely reactive ones. Preceding factors (that is, carryover effects) were also ruled out by analyses on the preceding shape presentation period (Fig. 2b). Furthermore, machine learning classification confirmed the spatial similarity of BOLD response, in both visual and somatosensory groups, between reactive sensory stimulation and anticipatory pre-stimulus decision periods (Fig. 3a,b) and the mutual dissimilarity of both from the preceding shape presentation period (Fig. 3c). Our results were also consistent with patterns of lateralized motor response, driven by the lateralized design in both sensory stimulation (left thumb press) and pre-stimulus decision (right finger press on button box) (Fig. 1e). Sensory domain-specific subcortical nuclei (LGN is visually sensitive; VPL is somatosensorily sensitive) were used as an additional validity check on our results, and both exhibited the same pattern of reactive and predictive response observed in the superior colliculus (Fig. 4).

In the human superior colliculus, the BOLD response to visual and somatosensory stimulation is organized across voxel depth. Prior non-human vertebrate work demonstrated that the superior colliculus is sensitive to both visual and somatosensory signals and that this sensitivity is organized around its layer structure. Visual stimuli elicit neural activity throughout the entire superior colliculus, but the concentration of visually-sensitive neurons is stronger in superficial layers, whereas somatosensory stimuli elicit neural activity exclusively in deep layers^{2,3,5}. Some prior evidence suggested that this layer-based distinction was present in humans (for example, dorsal superior colliculus showed greater BOLD response during oculomotor behaviors, and ventral superior colliculus showed greater BOLD response during visually guided somatomotor behaviors)^{10,11,34}. However, the limited resolution of this prior human research (collected at 3 Tesla) left it unclear whether the results (especially in deep layers) were driven by the superior colliculus itself, by neighboring CSF³⁵ or by adjacent midbrain structures that could not be easily distinguished at lower spatial resolutions (for example, PAG). The present work addressed these problems by using (1) 1.1-mm isotropic spatial resolution,

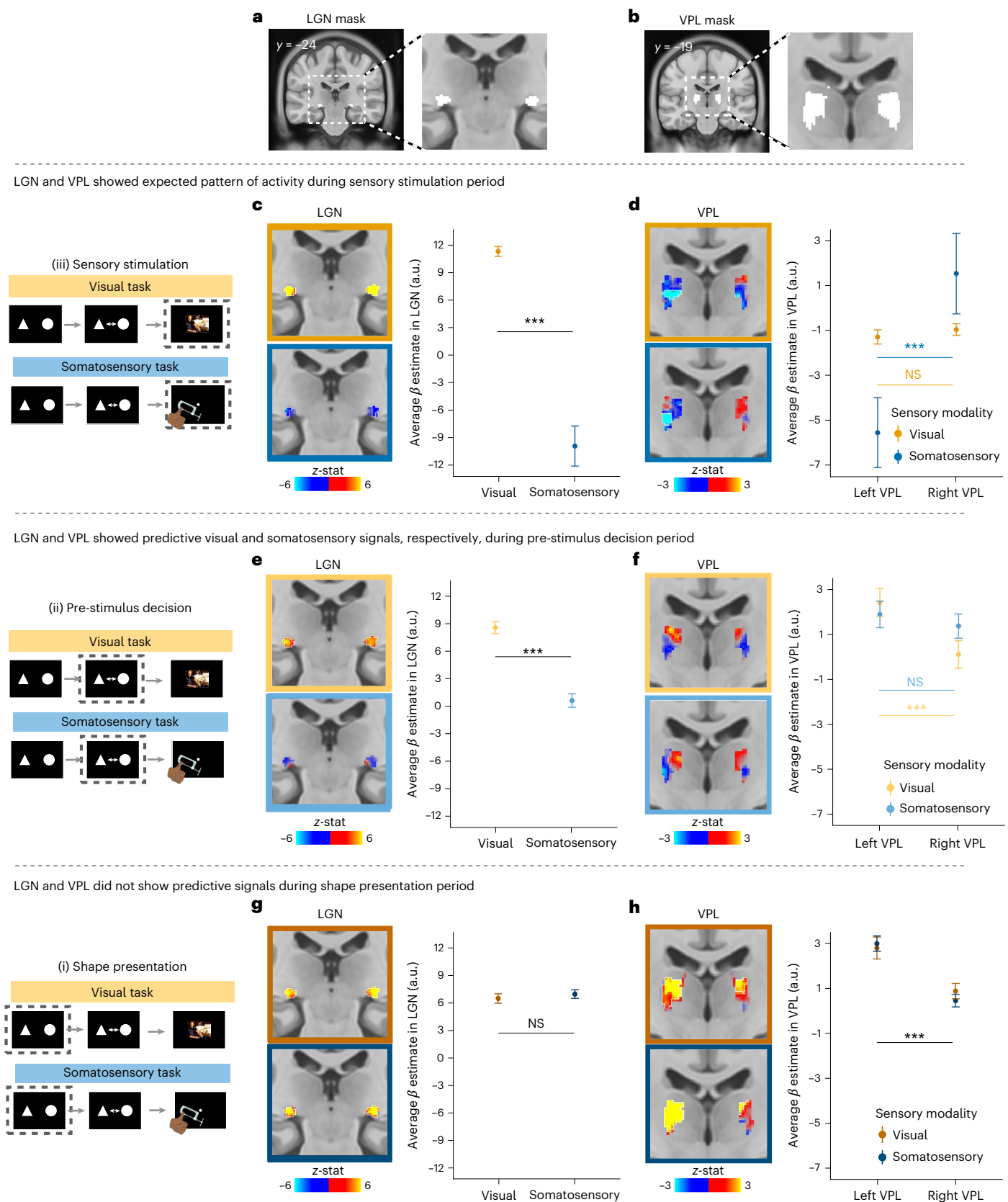
Fig. 4 | BOLD response in LGN and VPL thalamic nuclei was consistent with the sensory modality of subsequent sensory stimulation during pre-stimulus decision period but not during shape presentation period. a, b, LGN and VPL anatomical masks. **c**, During the sensory stimulation period, LGN BOLD response was greater for subjects receiving visual stimulation compared to subjects receiving somatosensory stimulation (main effect of sensory modality from linear mixed-effect model, $F_{1,74.32} = 54.20$, $P < 0.001$). **d**, In the sensory stimulation period, BOLD response in right VPL was greater than BOLD response in left VPL (two-sided paired t -test: $t_{39} = -4.75$, $P < 0.001$), consistent with the lateralized delivery of somatosensory pressure to the left thumb. Bilateral VPL showed no significant difference during visual stimulation (two-sided paired t -test: $t_{39} = -0.93$, $P = 0.358$; side $\times$ sensory modality interaction from linear mixed-effect model, $F_{1,1027.31} = 12.19$, $P < 0.001$). **e**, In the pre-stimulus decision period, LGN showed a similar pattern of relative BOLD response as during sensory stimulation, showing greater BOLD response in subjects anticipating visual stimulation versus subjects anticipating somatosensory stimulation (main effect of sensory modality from linear mixed-effect model, $F_{1,65.11} = 36.33$, $P < 0.001$). **f**, In the pre-stimulus decision period, VPL showed a left-lateralized BOLD response when anticipating visual stimulation (two-sided paired t -test: $t_{39} = 4.88$, $P < 0.001$) and a bilateral response when anticipating somatosensory stimulation

(two-sided paired t -test: $t_{39} = 1.06$, $P = 0.296$; side $\times$ sensory modality interaction from linear mixed-effect model, $F_{1,1137.99} = 7.14$, $P = 0.008$). We interpret the visual result as a contralateral response to the right-handed button press during the pre-stimulus decision period and the somatosensory result as both the contralateral left-lateralized response to the present right-handed button press as well as a right-lateralized anticipatory response to the upcoming somatosensory stimulation to the left thumb. **g**, In the shape presentation period, LGN showed no significant difference between visual and somatosensory groups (main effect of sensory modality from linear mixed-effect model, $F_{1,78.08} = 0.23$, $P = 0.636$). **h**, In the shape presentation period, VPL showed a left-lateralized response, which we interpret as anticipating and preparing the right-handed motor response in the upcoming pre-stimulus decision period (main effect of side from linear mixed-effect model, $F_{1,70.27} = 84.80$, $P < 0.001$). Colored voxels refer to z -statistics (task period > baseline). White outlines are superimposed onto clusters that passed a voxel-wise cutoff of $q < 0.05$ (one-tailed, FDR corrected) within the LGN or VPL masks. Error bars indicate ± 1 s.e. across subjects. Linear mixed-effect models or two-sided paired t -tests, as specified per panel: *** $P < 0.001$; NS, not significant. For all panels, $n = 40$ subjects per modality group (visual and somatosensory; 80 total).

(2) echo planar imaging (EPI)-based anatomical reconstruction to match anatomical and functional distortion and improve alignment³⁶ and (3) a preprocessing pipeline to identify and exclude PAG and CSF on a subject-specific basis, before conducting secondary alignment of all subjects to a common space for group analysis. These improvements

strengthen the evidence that the superficial–deep gradient in visual and somatosensory sensitivity is a feature common to both human and non-human primates.

In the human superior colliculus, both reactive and anticipatory BOLD responses to sensory stimulation are similarly organized across



voxel depth. To our knowledge, no prior work has observed that the human superior colliculus shows an anticipatory (that is, predictive) BOLD response to upcoming visual and somatosensory stimulation. However, evidence from non-human primate work is consistent with this observation. For example, neurons in the superficial layers of the superior colliculus are organized into retinotopic maps³⁷, and some of these neurons exhibit an anticipatory response when an impending saccade is expected to bring a visual stimulus into its receptive field^{25,38,39}. Furthermore, the superior colliculus may coordinate predictive signals in advance of motor commands: When a saccade is initiated, corollary discharges from the superior colliculus can stabilize visual input⁴⁰ or modulate visual processing to ensure perceptual continuity during rapid eye movements⁴¹. This prior work focused on predictive signals in the context of visuomotor integration in the superior colliculus, but our work provides additional evidence showing a reliable layer-dependent distinction based on anticipated visual or somatosensory input.

This anticipatory BOLD response is particularly relevant to allostatic and predictive processing accounts of brain function. Allostatic accounts emphasize that a brain organized around reaction is poorly suited to regulate its body in the context of a complex and ever-changing external environment. That is, an allostatic system should learn to anticipate and prepare for likely sensory consequences in advance of their arrival at sensory surfaces^{13–17}. Predictive processing accounts propose that top-down processing across the brain, in large part, predicts sensory input and filters it, so that ascending bottom-up signaling can be reduced to encoding prediction error^{18–20}. In the superior colliculus, corrective motor commands that follow in the wake of prediction error are a critical aspect of adaptive motor control^{42–44}. But such prior accounts of adaptive motor control have emphasized the role of the superior colliculus as a reactive coordinator of fast motor responses (for example, saccades) to sensory feedback rather than positioning it within the predictive processes. By contrast, the anticipatory response reported above suggests that the superior colliculus receives and carries predictive signals as part of the broader predictive cascade initiated before sensory stimulation arrives. As noted above, some retinotopic work anticipated this result^{25,38,39}, but the present work validates these earlier visual findings in humans and extends them into the somatosensory domain.

Some potential confounds should be noted. For instance, the anticipatory response in deep superior colliculus (and VPL; see below) could be driven by passive thumb motion induced by mechanical pressure on the thumb^{4,7,45,46}, which means that the results could stem from either somatosensory input or motor commands. Note, however, that this confound does not undermine the finding that visual and somatosensory stimulation elicit distinct patterns of reactive and anticipatory BOLD response along the superior colliculus layer gradient. The confound only leaves it ambiguous whether the BOLD response during anticipated somatosensory stimulation was driven by mechanical pressure on the thumb or by the motor commands that the pressure is inherently coupled with. Moreover, our resolution does not allow us to determine whether the observed deep-layer activity reflects a localized representation of the stimulated thumb or a more general response related to anticipatory/imminent aversive or alerting processes. Thus, in the case of somatosensory stimulation, although we interpret the effect in terms of anticipatory sensory prediction, we cannot rule out the possibility that it partly reflects non-specific upregulation in anticipation of a potentially unpleasant stimulus. Relatedly, intermediate/deep-layer activity during visual stimulation may reflect engagement of visuomotor circuitry in the superior colliculus (for example, motor preparation/orienting) rather than purely sensory visual processing in deeper layers. In addition, signals from adjacent structures (for example, PAG) may exert partial volume effects on the superior colliculus. However, this is unlikely to fully account for the observed effects, given both our subject-specific PAG/CSF masking (Methods) and our observations that, across task periods, the

correspondence of PAG–superior colliculus signal is inconsistent at their boundary (for PAG control analysis, see Supplementary Note 3 and Supplementary Figs. 2 and 3). Finally, because eye movements were not recorded during scanning, we acknowledge that we cannot definitively rule out subtle modality-dependent differences in gaze behavior as a contributing factor (for out-of-scanner eye-tracking experiment, see Supplementary Note 2). Future work is necessary to elaborate on these initial results, but the present work suggests that the superior colliculus does more than just react to prediction error and may be involved in a predictive cascade that begins before visual or somatosensory stimulation occurs.

Anticipatory BOLD responses to visual and somatosensory stimulation are present, respectively, in LGN and VPL thalamic nuclei. Like the superior colliculus, thalamic nuclei have typically been assessed by probing their reactive response to sensory stimuli. The LGN is responsive to experimental tasks involving early visual processing and visuomotor coordination⁴⁷, and the VPL is responsive to tasks probing early somatosensory processing⁴⁸. In the present work, we observed similar patterns of sensory-specific anticipatory BOLD responses in LGN, VPL and superior colliculus (Fig. 4). This convergence across multiple sensory structures suggests that anticipatory signals are distributed across a broader neural circuit, consistent with extensive evidence showing that the cortex sends descending projections to subcortical structures (including thalamic nuclei and the superior colliculus), enabling coordinated prediction across distributed sensory pathways^{33,49}. Future work is necessary to clarify the directionality and specificity of these predictive signals across the network.

The present work confirmed that the human superior colliculus is a layer-organized structure that processes both visual and somatosensory information and further demonstrated that the superior colliculus showed layer-specific patterns of BOLD response both in reaction to and in anticipation of visual and somatosensory stimuli. This finding is consistent with allostatic^{13–17} and predictive^{18–20} processing theoretical frameworks, which propose that the brain uses predictable sensorimotor contingencies to regulate adaptive behavior in complex environments. Future work can refine these results further and would particularly benefit from incorporating predictive contingencies across trials (or within trials, as in the present work). In sum, the present work demonstrates that subcortical nuclei are not passive recipients of sensory input but, rather, that they both react to and anticipate sensory input from the external world.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41593-026-02480-0>.

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Methods

Subjects

The final sample included 80 subjects ($M_{\text{age}} = 26.58$ years, s.d. = 5.87 years, 33 females, 47 males), with 40 completing the visual modality task and 40 completing the somatosensory modality task. At screening, subjects were excluded if they were pregnant, claustrophobic or had metal implants that could pose a risk during MRI scanning. Eligible subjects were recruited from the greater Boston area; were 18–40 years old, right handed and fluent English speakers with normal or corrected-to-normal vision; and had no known neurological or psychiatric disorders. Initially, 140 subjects were recruited, of whom 41 withdrew participation or ended the scan session before starting the present task. Of the remaining 99 subjects, 19 were excluded because all their runs contained poor-quality data—either excessive motion (>0.5 -mm frame-wise displacement in $>20\%$ of run repetition times) or other errors precluding analysis (for example, registration failures). In addition to the 19 subjects fully excluded, 41 other runs were excluded for poor quality, leaving 167 runs for the visual modality task and 167 runs for the somatosensory modality task (on average, 4.175 runs per subject). All subjects provided written informed consent and were compensated in accordance with the guidelines set by the institutional review board of Massachusetts General Hospital.

Experimental paradigm

The paradigm involved a reinforcement learning task based on previous work⁵⁰, in which subjects associated visual cues with aversive or neutral stimuli. Subjects completed five runs with 24 trials in each, and each subject received either visual or somatosensory stimulation (but not both). Each trial involved a sequence of shape presentation, pre-stimulus decision and sensory stimulation periods (Fig. 1a). Subjects were shown pairs of shapes to select between and were instructed (1) that shapes had a probability of leading to aversive sensory feedback (that is, unpleasant images or strong pressure on the thumb), (2) that this probability could change over time and (3) that they should avoid shapes that led to aversive outcomes. The probabilities of aversive reinforcement were constrained between 20% and 80% (mean likelihood = 50%) and determined by pre-computed random walks (four walks were pseudorandomly balanced across subjects). Analyses of task periods were averaged across trials and runs, making the influence of probabilistic reinforcement irrelevant to our results. The shape presentation period presented both shapes (circle and triangle) with a jittered presentation duration (1.5–5.5 seconds, 3.5 seconds on average). Next, the pre-stimulus decision period (2 seconds, not jittered) signaled that a shape would be selected. In 50% of trials, subjects were shown a double-sided white arrow and used the button box in their right hand to select a shape, at which point the arrow became a one-sided yellow arrow to indicate their choice (participant decision). In the other 50% of trials, a yellow one-sided arrow appeared to indicate which shape was selected randomly by the computer (computer decision), ensuring that subjects were exposed to the outcome of both shapes. Next, after a jittered inter-trial interval (ITI) (1–5 seconds, 3 seconds on average), the sensory stimulation period occurred (3 seconds), followed by another jittered ITI (sampled from Poisson distribution, 6.5–18.5 seconds, 12.5 seconds on average).

Visual stimuli were presented on an 8.5" × 15" (21.59 × 38.10 cm) monitor, approximately 32" (81.28 cm) from the subject's eyes, accounting for mirror reflection on the head coil. During the shape presentation and pre-stimulus decision periods, the displayed triangle measured 2.5" × 2.5" (6.35 × 6.35 cm), corresponding to a visual angle of 4.5° × 4.5°, and the circle had a diameter of 2.5" (6.35 cm), corresponding to a visual angle of 4.5°. The positions of the triangle and circle were counterbalanced across left and right hemifields on each trial. For the visual task, during the sensory stimulation period, the images measured approximately 5.5" × 4.13" (13.97 × 10.50 cm) and were presented at the center of the screen, corresponding to a visual angle of 9.8° × 7.3°.

Across trials, both visual and somatosensory stimulation periods presented neutral and aversive stimuli. During visual stimulation, subjects were shown images from a pre-selected set of IAPS images²⁸, normed in an independent sample⁵¹ on valence (that is, positive versus negative feeling) and arousal (that is, feelings of high/low physiological activation). During somatosensory stimulation, mechanical pressure was applied at light (neutral condition; approximately 3 kg cm⁻²) or aversive (approximately 5 kg cm⁻²) intensities. These benchmark intensities were selected based on previous studies^{52,53} and were intended to induce low-intensity pain. The position of the pressure device was adjusted on each subject's thumb before scanning to ensure that the two levels were felt as distinct and that neither was severely uncomfortable. Despite these steps, subjects rated the two levels as equivalently painful on average, and so distinctions between neutral and aversive visual and somatosensory stimuli were not used in our primary analyses. Valence and arousal norms, subjects' own ratings and superior colliculus depth-dependent responses separated by stimulus intensity are reported in Supplementary Note 4 and Supplementary Fig. 1.

fMRI data acquisition and preprocessing

Gradient-echo EPI BOLD fMRI was performed on a 7-Tesla Siemens MRI scanner. Functional images were acquired using GRAPPA-EPI sequence: echo time = 28 ms, repetition time = 2.34 seconds, flip angle = 75°, slice orientation = transversal (axial), anterior to posterior phase encoding, voxel size = 1.1 mm isotropic, gap between slices = 0 mm, number of slices = 123, field of view = 205 × 205 mm², GRAPPA acceleration factor = 3, echo spacing = 0.82 ms, bandwidth = 1,414 Hz per pixel and partial Fourier in the phase encode direction = 7/8. A custom-built 32-channel radiofrequency coil head array was used for reception. Radiofrequency transmission was provided by a de-tunable band-pass birdcage coil.

Structural images were acquired using a T1-weighted (T1w) EPI sequence, selected so that functional and structural images had similar spatial distortions, which facilitated co-registration and subsequent normalization of data to MNI space³⁶. Structural scan parameters were as follows: echo time = 22 ms, repetition time = 8.52 seconds, flip angle = 90°, number of slices = 126, slice orientation = transversal (axial), voxel size = 1.1 mm isotropic, gap between slices = 0 mm, field of view = 205 × 205 mm², GRAPPA acceleration factor = 3, echo spacing = 0.82 ms, bandwidth = 1,414 Hz per pixel and partial Fourier in the phase encode direction = 6/8.

fMRI data were preprocessed using the fMRIPrep pipeline version 1.1.2 (ref. 54), a Nipype-based tool⁵⁵. Full pipeline details can be found at <https://fmripipeline.org/en/1.1.2/workflows.html>. Spatial normalization of anatomical data to the ICBM 152 Nonlinear Asymmetrical template version 2009c was performed through nonlinear registration with the antsRegistration tool of ANTs version 2.1.0 (ref. 56), using brain-extracted versions of both T1w volume and template. Brain tissue was segmented into CSF, white matter and gray matter (FAST, FSL version 5.0.9)⁵⁷. Functional data were slice time corrected (dTshift, AFNI 16.2.07)⁵⁸, motion corrected (MCFLIRT)⁵⁹ and co-registered to the T1w using boundary-based registration⁶⁰ with six degrees of freedom (FLIRT, FSL version 5.0.9)⁵⁹. Motion-correcting transformations, BOLD-to-T1w transformation and T1w-to-template (MNI 2009c asymmetric) warp were concatenated and applied in a single step using antsApplyTransforms in ANTs version 2.1.0 using Lanczos interpolation. Physiological noise regressors were extracted using the aCompCor method⁶¹, taking the top five principal components from subject-specific CSF and white matter masks. CSF and white matter masks were thresholded at 99% probability, with the CSF mask constrained to the ventricles (using the ALVIN mask)⁶² and eroding the white matter mask using the binary erosion function in SciPy version 1.6. Frame-wise displacement⁶³ was calculated for each functional run using the implementation in fMRIPrep version 1.1.2. For all subjects, the quality of brain masks, tissue segmentation and MNI registration was visually inspected for errors using the html figures provided by the fMRIPrep pipeline.

Superior colliculus localization and alignment

Subject alignment to MNI space is not optimized to align brainstem structures such as the superior colliculus³⁵, and so second-pass alignment was performed to bring subject-specific superior colliculus ROIs into a common space for group analysis (for a similar approach, see refs. 30,64,65). First, anatomical data in MNI 2009c asymmetric space were used to create a hand-drawn superior colliculus mask identifying candidate voxels for inclusion (available at <https://neurovault.org/collections/LCUVQSUL/>). Next, in each subject, this hand-drawn mask was further refined by applying subject-specific masks, excluding voxels in a CSF mask (>50% probability) and including voxels in gray matter and white matter masks (>25% probability). Subject-specific masks were further refined by excluding voxels in subject-specific masks of the PAG, which abuts deep layers of the superior colliculus (for details on the PAG mask, see refs. 64,65). Finally, to guard against partial volume effects from the nearby cerebral aqueduct, the subject-specific masks were refined by masking any voxels producing large residuals (>85th percentile) in the subject-level univariate general linear model (see univariate analysis below) and cleaned again by removing any remaining small clusters (<10 voxels, 13.31 mm³). For all subjects (that is, both visual and somatosensory groups), subject-specific superior colliculus masks were warped from MNI space to a secondary common space using DARTEL³², and the subject-specific transformations were applied to each subject's generalized linear model (GLM) contrast estimates to align them to a common space for group analysis. The success of this strategy was assessed by comparing Dice coefficients across subject-specific superior colliculus masks before and after alignment, and we observed that DARTEL alignment substantially increased Dice coefficients from before ($M_{\text{before}} = 89.53\%$, s.d. = 3.09%) to after ($M_{\text{after}} = 94.61\%$, s.d. = 2.16%) the DARTEL warp. After thresholding at 50% probability, the average of the warped subject-specific superior colliculus masks was used as a group mask in all lateral analyses (consistent with refs. 64,65). Control analyses are reported in the Supplementary Information: between-group comparisons of motion, signal-to-noise ratio and percent signal change (Supplementary Note 5 and Supplementary Fig. 4); validation of regressor separability and canonical hemodynamic response function (HRF) timing (Supplementary Notes 6 and 7); and an out-of-scanner eye-tracking experiment addressing gaze differences across task periods (Supplementary Note 2).

Superior colliculus layer estimation

In the group-aligned superior colliculus mask, voxel-wise layer depth (voxel depth) was calculated by measuring Euclidean distance to an anchor point (manually identified by author D.C.) ventromedial to each side of the superior colliculus (right MNI coordinates: $x = 3$, $y = -30$, $z = -10$; left MNI coordinates: $x = -3$, $y = -30$, $z = -10$; for a related approach, see refs. 66,67). These Euclidean distances were normalized to assign each superior colliculus voxel in the group-aligned mask a voxel depth score from 0 (closest to the anchor; that is, deep layers) to 1 (farthest from the anchor; that is, superficial layers). Superior colliculus mask and layer estimation are available online (<https://neurovault.org/collections/LCUVQSUL/>). For detailed measurements of the superior colliculus and its layer-specific sizes, see Supplementary Note 8 and Supplementary Table 1. For post hoc validation of this layer estimation, see Supplementary Note 9.

LGN and VPL masks

LGN mask was acquired through the Brainstem Navigator toolkit (<https://www.nitrc.org/projects/brainstemnavigator/>)^{68,69}. VPL mask was acquired through the CANlab combined atlas (https://github.com/canlab/Neuroimaging_Pattern_Masks/tree/master/Atlases_and_parcellations/2018_Wager_combined_atlas)^{70,71}.

Whole-brain univariate GLM

First-level GLMs (FSL version 5.0.9) were conducted within each run in each subject and modeled the effects, relative to fixation, of (1)

shape presentation, (2) pre-stimulus decision (modeling participant and computer decision types as separate regressors) and (3) sensory stimulation periods (modeling aversive and neutral stimulations as separate regressors). Durations of each regressor are as reported above (experimental paradigm; Fig. 1a), with the exception of somatosensory stimulation, where the onset of pressure (0–1 seconds, 1-second duration) was modeled separately from the rest of the pressure stimulation (1–3 seconds, 2-second duration), consistent with previous studies of pain stimulation using similar experimental design⁵⁰. This subdivision of task period is protocol driven, and primary inferences do not rely on dissociating these phases (see Supplementary Fig. 5 for comparison of the 1-second onset model versus the full 3-second duration models). All regressors were convolved with a double-gamma hemodynamic function. GLMs included additional terms modeling the run intercept, 24 motion terms (that is, six translation/rotation terms, their mean-centered squares, derivatives and squared derivatives), 10 aCompCor components⁶¹, a discrete cosine transformation set with a minimum period of 120 seconds and spike regressors (for repetition times with >0.5-mm frame-wise displacement)⁷². A small subset of trials (0.6% of all trials) was also removed with spike regressors, due to a programming error that made it impossible to reconstruct pre-stimulus decision period timing when subjects did not choose a shape. Seven runs were dropped from analysis if more than 20% of repetition times were excluded via spike regressors.

Run-level contrasts were warped using DARTEL (see Superior colliculus localization and alignment section). For whole-brain analyses (Extended Data Fig. 1), the data were then smoothed with a 3-mm FWHM across the entire brain. To minimize interference of signals from nearby structures, for ROI analyses (Figs. 1d, 2a,b and 4), the data were smoothed using a 3-mm FWHM within group-aligned superior colliculus, LGN or VPL masks, using 3dBlurInMask in AFNI version 16.2.07 (ref. 58). The data were then averaged across runs within each subject to produce subject-level contrast maps. To estimate whole-brain group-level effects, subject-level contrast maps were submitted to a GLM (ordinary least squares; one-sample *t*-test; Extended Data Fig. 1) and thresholded using a false discovery rate ($q_{\text{FDR}} < 0.01$) and a minimum cluster extent of 20 voxels (26.62 mm³). In volumetric results within ROIs (Figs. 1d, 2a,b and 4), unthresholded *z*-statistic maps are shown with white outline indicating voxel clusters that passed the threshold of $q_{\text{FDR}} < 0.05$, and no minimum voxel extent was applied. Unthresholded *z*-statistics maps are available online (<https://neurovault.org/collections/LCUVQSUL/>).

Mixed-effects analysis

For analyses within the superior colliculus (Figs. 1e,f and 2), unsmoothed run-level whole-brain contrasts were masked by the group-aligned superior colliculus mask and submitted to a mixed-effects analysis (lme4 (ref. 73) and Satterthwaite degrees of freedom, implemented in lmerTest⁷⁴ in R⁷⁵). For descriptive post hoc paired comparisons reported in the 'Results', β estimates were first averaged across runs within each subject (one value per subject), and two-sided paired *t*-tests were then computed across subjects within each sensory modality group. Superior colliculus voxel depth was categorized into 10 ordinal bins for the purpose of analysis, and each voxel was tagged according to its categorical voxel depth and side (that is, left/right). Run-level averages were calculated within these categories. Code-based summaries of all models are available in Extended Data Table 1.

For the mixed-effects model of the sensory stimulation period, fixed main effects included side (left/right), voxel depth (10 categories, spanning deep to superficial poles), sensory modality (visual/somatosensory) and fixed-effect interactions including sensory modality $\times$ side and sensory modality $\times$ voxel depth (remaining main effects and interactions were either uninterpretable or not of theoretical interest and so were omitted from the model). The sensory stimulation model included a by-subject random intercept and random slopes for all fixed effects listed above (except for the interactions

involving sensory modality, which was a between-subjects variable). The mixed-effects models of the pre-stimulus decision period and the shape presentation period were both identical to the sensory stimulation period where sensory modality was labeled using the upcoming modality of sensory stimulation; the pre-stimulus decision period model additionally included terms for decision type.

VPL and LGN models in each period were identical to the superior colliculus model, with the exception that they did not include voxel depth. Unsmoothed run-level contrasts masked by either VPL or LGN masks were used for these analyses.

Multivariate pattern analysis

In sensory stimulation and pre-stimulus decision periods, current or subsequent visual and somatosensory stimulation produced similar patterns across layers in the superior colliculus, whereas, in the shape presentation period, a different pattern appeared to be present. The voxel-wise similarity of these patterns in the superior colliculus was empirically assessed using linear SVM classifiers⁷⁶ implemented in the SVM function in scikit-learn⁷⁷. Classifiers were trained on all voxels in the aligned group-level superior colliculus mask and attempted to predict whether a test image came from a subject receiving visual or somatosensory stimulation, after being trained on patterns in contrast images during sensory stimulation, pre-stimulus decision and shape presentation periods. Critically, the visual/somatosensory stimulation that differentiated the two subject groups was delivered only during the sensory stimulation period, whereas sensory input during earlier pre-stimulus and shape presentation periods was equivalent between visual and somatosensory conditions, making it an open question whether classifiers trained on sensory stimulation contrasts could identify other condition contrasts as coming from subjects receiving visual or somatosensory stimulation.

Samples used in the classifier were unsmoothed first-level contrast maps for each run, in each subject. Because subjects completed multiple runs, each subject had multiple run-level observations (1–5 runs; Methods). Classifiers were trained using leave-one-subject-out cross-validation (that is, 80 folds), with each fold training on data from 79 subjects and testing on the remaining subject (for whom all runs were excluded from training and used only for testing). In each fold, all runs from the held-out subject were excluded from training and used only for testing. Out-of-sample classification was assessed by applying the trained classifier from one period (for example, sensory stimulation; 79 subjects) to the corresponding data of the held-out subject from another period (for example, pre-stimulus decision period; the remaining subject). Fold-wise accuracy was calculated across the held-out subject's run-level observations (meaning a subject with five runs could produce accuracies of 0, 0.2, 0.4, 0.6, 0.8 or 1).

Classification accuracy was evaluated using a non-parametric permutation test where subject-level (not run-level) visual/somatosensory labels were shuffled for 1,000 iterations to create a null distribution (that is, the permuted label was applied to all runs from the same subject). For each permutation iteration, the full leave-one-subject-out procedure was repeated. Within each permutation, all fold-wise accuracies were averaged to obtain a single permuted mean accuracy. Across the 1,000 permutations, this yielded a null distribution of 1,000 permuted mean accuracies. One-sided *P* values were calculated by taking the proportion of permuted mean accuracies in the null distribution that were greater than or equal to the observed mean accuracy. This analysis was also repeated using a balanced paired leave-two-subjects-out cross-validation scheme (holding out one visual and one somatosensory subject per fold), ensuring that both training sets contained *n* = 39 subjects per modality. Results were similar and are reported in Supplementary Note 10 and Supplementary Table 2.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Raw and preprocessed data can be found at <https://openneuro.org/datasets/ds008447/versions/1.1.0>.

Code availability

Analysis codes can be found at <https://github.com/danlei-chen/prediction-in-Superior-Colliculus-7T>.

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Author contributions

P.A.K., L.L.W., M.B., T.D.W., K.S.Q., A.B.S. and L.F.B. conceptualized the study and designed the experiment. D.C., Z.K., P.A.K., J.-K.C., J.Z. and J.E.T.-B. performed the experiment. D.C., M.B.-Z., P.A.K., P.W.S. and

J.E.T.-B. analyzed the experiment. D.C., L.F.B. and J.E.T.-B. wrote the first draft of the manuscript, after which all authors contributed to its editing.

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Competing interests

The authors declare no competing interests.

Additional information

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Correspondence and requests for materials should be addressed to Danlei Chen, Lisa Feldman Barrett or Jordan E. Theriault-Brown.

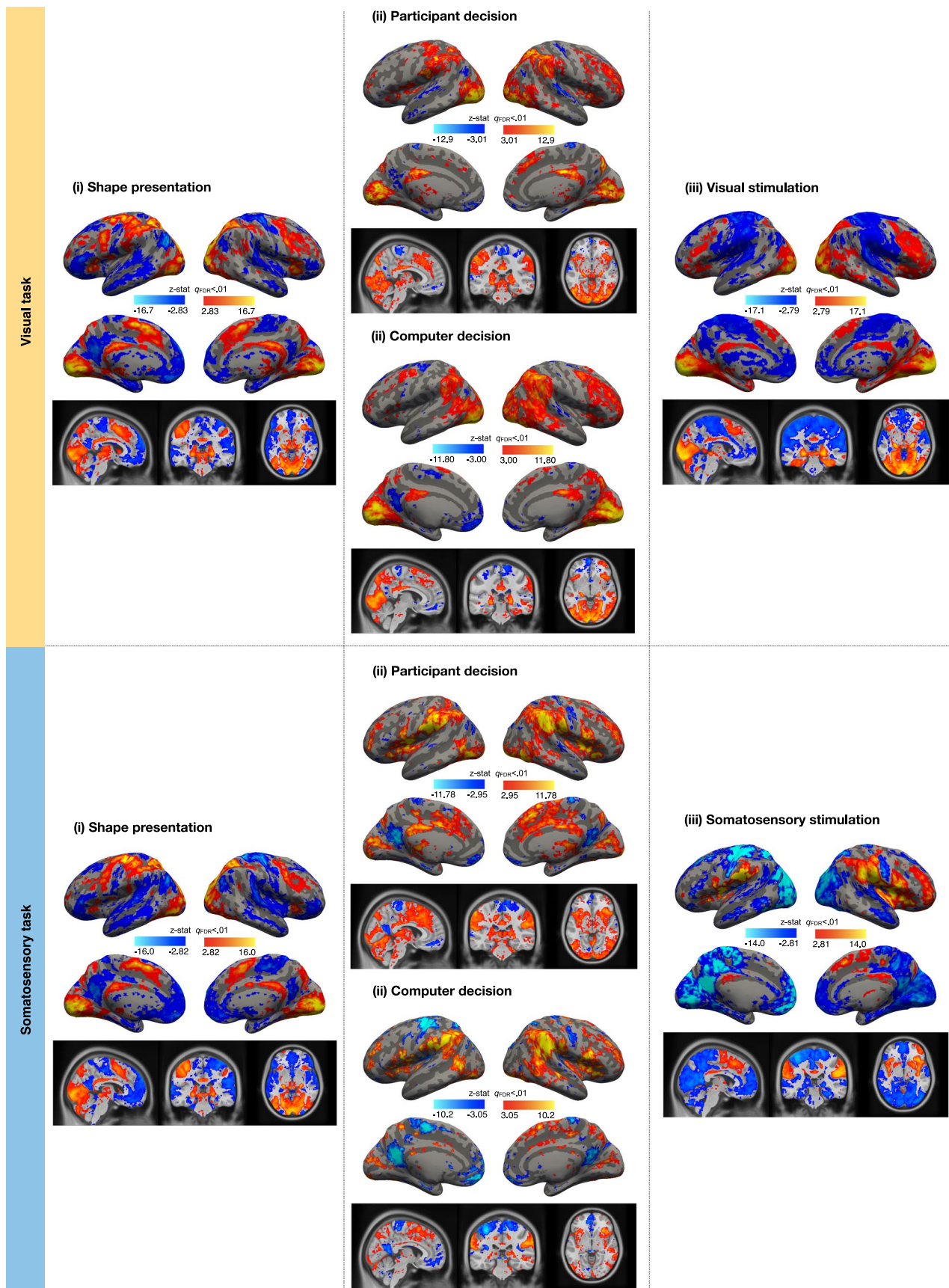
Peer review information *Nature Neuroscience* thanks Dietrich Schwarzkopf and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.

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Extended Data Table 1 | ROI analysis results of linear mixed effect models using Type III Analysis of Variance (ANOVA) during all task-related periods for (A) superior colliculus, (B) LGN, and (C) VPL

Region	Task period	Effect	F	df	P
(A) Superior colliculus	Sensory stimulation	voxel depth	23.52	(1, 65.79)	7.9×10^{-6}
		side	14.92	(1, 67.29)	2.5×10^{-4}
		sensory modality	26.88	(1, 70.04)	2.0×10^{-6}
		voxel depth : sensory modality	30.17	(1, 65.79)	6.9×10^{-7}
		side : sensory modality	9.37	(1, 67.29)	0.003
	Pre-stimulus decision	voxel depth	12.93	(1, 66.76)	6.2×10^{-4}
		side	1.59	(1, 72.19)	0.211
		decision type	7.00	(1, 74.99)	0.010
		sensory modality	0.01	(1, 72.78)	0.930
		voxel depth : sensory modality	30.88	(1, 66.76)	5.2×10^{-7}
		side : sensory modality	1.95	(1, 72.19)	0.167
		decision type : sensory modality	2.12	(1, 74.99)	0.150
	Shape presentation	voxel depth	4.29	(1, 61.68)	0.043
		side	4.90	(1, 75.00)	0.030
		sensory modality	2.47	(1, 72.15)	0.120
		voxel depth : sensory modality	6.92	(1, 61.68)	0.011
		side : sensory modality	0.04	(1, 75.00)	0.837
(B) LGN	Sensory stimulation	sensory modality	54.20	(1, 74.32)	2.0×10^{-10}
		side	0.68	(1, 1196.51)	0.408
		sensory modality : side	0.64	(1, 1196.51)	0.424
	Pre-stimulus decision	decision type	19.21	(1, 45.78)	6.8×10^{-5}
		side	1.02	(1, 1098.55)	0.313
		sensory modality	36.33	(1, 65.11)	8.6×10^{-8}
		decision type : side	0.42	(1, 1098.55)	0.515
		decision type : sensory modality	0.90	(1, 45.78)	0.348
		side : sensory modality	0.33	(1, 1098.55)	0.566
		decision type : side : sensory modality	0.35	(1, 1098.55)	0.553
	Shape presentation	sensory modality	0.23	(1, 78.08)	0.636
		side	30.94	(1, 320.42)	5.6×10^{-8}
		sensory modality : side	0.36	(1, 320.42)	0.549
(C) VPL	Sensory stimulation	sensory modality	1.43	(1, 66.95)	0.235
		side	20.00	(1, 1027.31)	8.6×10^{-6}
		sensory modality : side	12.19	(1, 1027.31)	5.0×10^{-4}
	Pre-stimulus decision	decision type	41.23	(1, 64.45)	1.9×10^{-8}
		side	16.23	(1, 1137.99)	6.0×10^{-5}
		sensory modality	0.26	(1, 69.40)	0.609
		decision type : side	16.52	(1, 1137.99)	5.1×10^{-5}
		decision type : sensory modality	0.39	(1, 64.45)	0.534
		side : sensory modality	7.14	(1, 1137.99)	0.008
		decision type : side : sensory modality	0.36	(1, 1137.99)	0.549
	Shape presentation	sensory modality	0.03	(1, 75.20)	0.870
		side	84.80	(1, 70.27)	1.1×10^{-13}
		sensory modality : side	1.67	(1, 70.27)	0.201

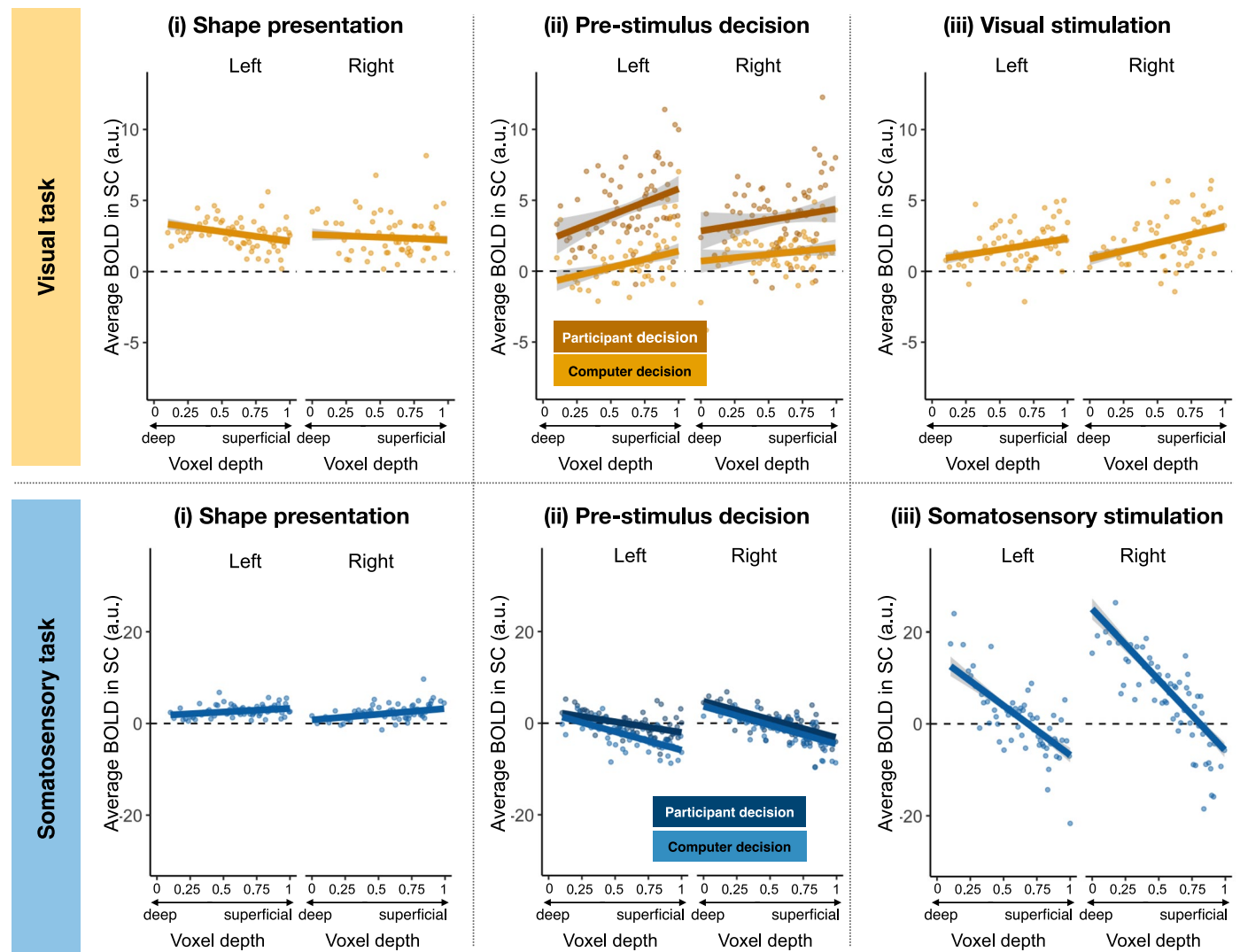
All analyses performed using the lme4 package in R, and degrees of freedom were estimated using Satterthwaite's method.



Extended Data Fig. 1 | Whole-brain contrasts of all task-related periods.

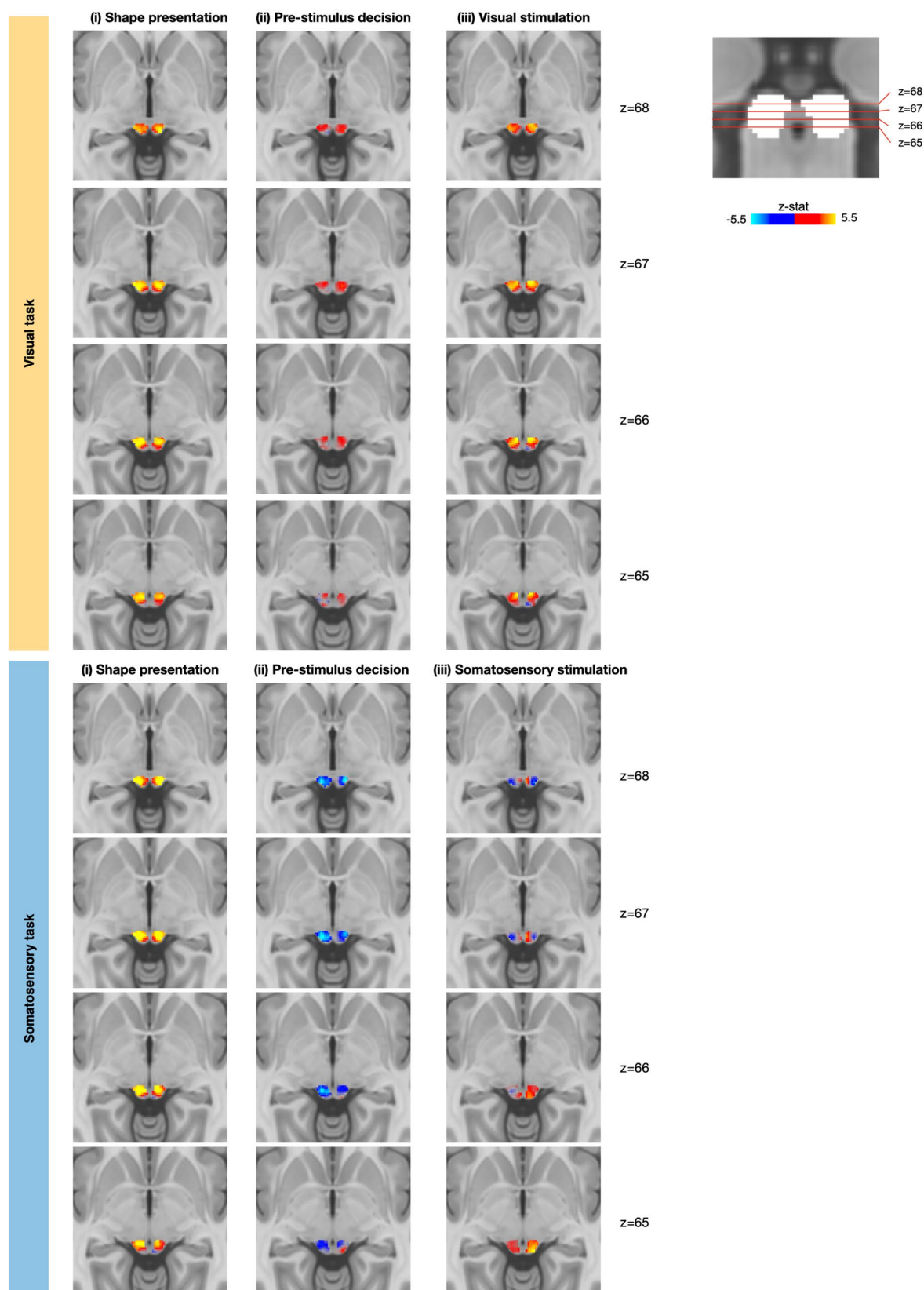
In both visual and somatosensory tasks, each task-related period was modeled for the shape presentation period, the pre-stimulus decision period (for both participant and computer decision types), and the sensory stimulation period.

All maps are z-statistics thresholded at $q < 0.01$ (FDR adjusted). 3 mm whole-brain smoothing was applied. For all panels, $n = 40$ subjects per modality group (visual, somatosensory; 80 total).

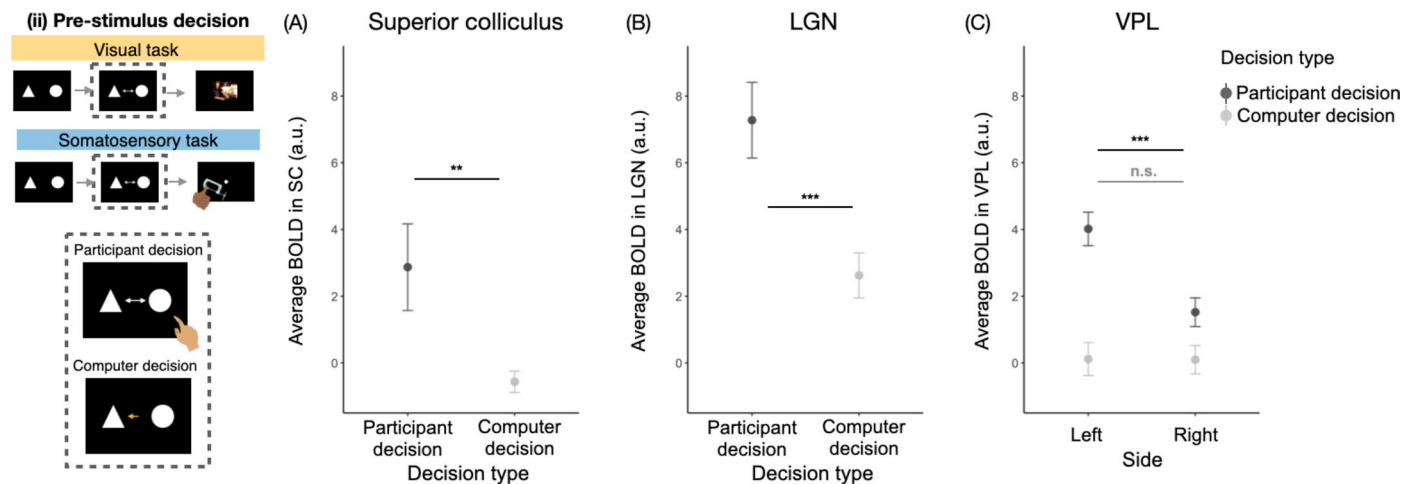


Extended Data Fig. 2 | Superior colliculus BOLD response across all task-related periods. Within the superior colliculus, BOLD signal was averaged across subject-level contrasts (y-axis) and was correlated with voxel depth estimates (x-axis) for each task-related period. Dots indicate mean BOLD response at

each depth bin. A linear relationship was estimated between voxel depth and amplitude of BOLD response. Shaded areas indicate 95% confidence interval. No smoothing was applied in data used in this figure. For all panels, $n = 40$ subjects per modality group (visual, somatosensory; 80 total). SC: superior colliculus.



Extended Data Fig. 3 | Superior colliculus activation maps during the shape presentation, pre-stimulus decision, and sensory stimulation periods. Axial slices are shown at $z = 65, 66, 67$, and 68 . All maps are z-statistics without thresholding. For all panels, $n = 40$ subjects per modality group (visual, somatosensory; 80 total).



Extended Data Fig. 4 | Movement-related BOLD response during the pre-stimulus decision period in superior colliculus, LGN, and VPL. (a) In the pre-stimulus decision period, superior colliculus showed greater BOLD response when subjects were instructed to make a button press (that is, participant decision) compared to when a decision was made for them by the computer (that is, computer decision). To test whether decision type modulated the depth-dependent superior colliculus signal, we fit a linear mixed-effects model including the decision_type:depth term for both fixed and random effects. The model shows no decision type $\times$ layer interaction ($F_{(1,68,11)}=1.64, P=0.205$) and no decision type $\times$ layer $\times$ modality interaction ($F_{(1,68,11)}=0.09, P=0.754$), suggesting that the depth-dependent pattern does not vary by decision type

(see also the lack of difference in layer-dependent activity between decision types in Extended Data Fig. 2). **(b)** LGN showed greater BOLD response when subjects were instructed to make a button press compared to when a decision was made for them by the computer. **(c)** The VPL showed left-lateralized BOLD response when subjects pressed a button to indicate their response but showed no lateralization when the computer decided for them, consistent with a motor-driven contralateral response in VPL. For all panels, $N=40$ subjects per modality group (visual, somatosensory; 80 total). Error bars indicate ± 1 standard error. Linear mixed-effect models or two-sided paired t-tests, as specified per panel: ** $P < 0.01$; *** $P < 0.001$; n.s., not significant. SC: superior colliculus.

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Data collection PsychoPy

Data analysis General linear models (whole-brain and ROI) were computed in FSL (version 5.0.9). Linear mixed-effects models were fit in R using the lme4 and lmerTest packages, with Satterthwaite degrees of freedom. Multivariate pattern analyses used linear support vector machine classifiers implemented in scikit-learn. Custom analysis code is available at <https://github.com/danlei-chen/prediction-in-Superior-Colliculus-7T>.

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio [guidelines for submitting code & software](#) for further information.

Data

Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our [policy](#)

Raw and preprocessed data can be found at <https://openneuro.org/datasets/ds008447/versions/1.1.0>.

Research involving human participants, their data, or biological material

Policy information about studies with [human participants or human data](#). See also policy information about [sex, gender \(identity/presentation\), and sexual orientation](#) and [race, ethnicity and racism](#).

Reporting on sex and gender	As this was a study of fundamental brain function across healthy individuals, we did not test the effects of sex or gender. We report information about participant's gender only to describe the characteristics of the sample. Participants reported gender (male, female) via self-report questionnaire, including 33 females and 47 males.
Reporting on race, ethnicity, or other socially relevant groupings	As this was a study of fundamental brain function across healthy individuals, we did not test the effects of any socially constructed categorization variables. See above for discussion of gender.
Population characteristics	All subjects were between 18 and 40 years old, were right-handed, had normal or corrected to normal vision, were not pregnant, were fluent English speakers, and had no known neurological or psychiatric illnesses. Subjects were excluded from participating in the study if they were claustrophobic or had any metal implants that could cause harm during scanning.
Recruitment	Participants were recruited from the greater Boston area via electronic advertisements and paper flyers posted in local establishments.
Ethics oversight	All subjects provided written informed consent and study procedures were completed as approved by the institutional review board of Massachusetts General Hospital.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	Target sample size was 160 for this project given power analysis for tasks not relevant to the current analyses. We stopped recruitment at 140 due to equipment technical issues and COVID. The final sample included 80 subjects, with 40 completing the visual modality task, and 40 completing the somatosensory modality task.
Data exclusions	Initially, 140 subjects were recruited, of which 41 withdrew participation or ended the scan session before starting the present task. Of the remaining 99 subjects, 19 were excluded because all their runs contained poor quality data—either excessive motion (> .5mm framewise displacement in > 20% of run TRs), or other errors precluding analysis (e.g., registration failures). In addition to the 19 subjects fully excluded, 41 other runs were excluded for poor quality, leaving 167 runs for the visual modality task, and 167 runs for the somatosensory modality task (on average, 4.175 runs per subject).
Replication	We observed a consistent pattern of superior colliculus layer activity across two independent participant samples during a task period of matched reactive and predictive sensorimotor demand.
Randomization	Participants were randomly assigned to one of two conditions for the task. When feasible, the experimenter attempted to balance participants across conditions by gender and age group during recruitment.
Blinding	Blinding was not possible because different experimental apparatus was used for visual and somatosensory groups.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☐	☒ MRI-based neuroimaging

Plants

Seed stocks	Report on the source of all seed stocks or other plant material used. If applicable, state the seed stock centre and catalogue number. If plant specimens were collected from the field, describe the collection location, date and sampling procedures.
Novel plant genotypes	Describe the methods by which all novel plant genotypes were produced. This includes those generated by transgenic approaches, gene editing, chemical/radiation-based mutagenesis and hybridization. For transgenic lines, describe the transformation method, the number of independent lines analyzed and the generation upon which experiments were performed. For gene-edited lines, describe the editor used, the endogenous sequence targeted for editing, the targeting guide RNA sequence (if applicable) and how the editor was applied.
Authentication	Describe any authentication procedures for each seed stock used or novel genotype generated. Describe any experiments used to assess the effect of a mutation and, where applicable, how potential secondary effects (e.g. second site T-DNA insertions, mosaicism, off-target gene editing) were examined.

Magnetic resonance imaging

Experimental design

Design type	task; event-related
Design specifications	Subjects completed a reinforcement learning task consist of 5 runs with 24 trials in each, and each subject received either visual or somatosensory stimulation (but not both). Each trial involved a sequence of shape presentation, pre-stimulus decision, and sensory stimulation periods (Fig. 1A). Participants were shown pairs of shapes to select between and were instructed that (a) shapes had a probability of leading to aversive sensory feedback (i.e., unpleasant images, or strong pressure on the thumb), (b) that this probability could change over time, and (c) that they should avoid shapes that led to aversive outcomes. The probabilities of aversive reinforcement were constrained between 20-80% (mean likelihood = 50%) and determined by precomputed random walks (four walks were pseudorandomly balanced across participants). Analyses of task periods were averaged across trials and runs, making the influence of probabilistic reinforcement irrelevant to our results. The shape presentation period presented the circle and the triangle and was jittered (1.5-5.5 sec., 3.5 sec. on average). Next, the pre-stimulus decision period (2 sec.) signaled that a shape would be selected. In 50% of trials, participants were shown a double-sided white arrow and used the button box in their right hand to select a shape, at which point the arrow became a one-sided yellow arrow to indicate their choice (participant decision). In the other 50% of trials, a yellow one-sided arrow appeared to indicate which shape was randomly selected by the computer (computer decision), ensuring subjects were exposed to the outcome of both shapes. Next, after a jittered inter-trial interval (1-5 sec., 3 sec. on average), the sensory stimulation period occurred (3 sec.), followed by another jittered ITI (sampled from Poisson distribution, 6.5-18.5 sec., 12.5 sec. on average).
Behavioral performance measures	Across trials, both visual and somatosensory stimulation periods presented neutral and aversive stimuli. During visual stimulation, subjects were shown images from a pre-selected set of IAPS images, normed in an independent sample on valence (i.e., positive vs. negative feeling) and arousal (i.e., feelings of high/low physiological activation). Consistent with our design, the set of images selected to provide negative feedback were more negatively valenced on a scale of 1-10 (Mvalence = 2.93; SD = .67) and more highly arousing (Marousal = 4.75; SD = .58), compared to the set of images selected to provide neutral feedback (valence: Mvalence = 5.52; SD = .57, $t(44) = 9.30$, $p < .001$; arousal: Marousal = 3.16; SD = .57, arousal, $t(44) = 13.72$, $p < .001$). During somatosensory stimulation, mechanical pressure was applied at light (neutral condition; approximately 3 kg/cm ²) or aversive intensities (approximately 5 kg/cm ²). These benchmark intensities were selected based on prior studies (e.g., Krishnan et al., 2016; López-Solà et al., 2017) and were intended to induce low intensity pain. The position of the pressure device was adjusted on each subject's thumb before scanning to ensure that the two levels were felt as distinct and that neither was severely uncomfortable. Analyses on valence and arousal ratings of the stimuli based on an initial practice scan were included in the Supplementary Material.

Acquisition

Imaging type(s)	structural and functional
Field strength	7 Tesla
Sequence & imaging parameters	Gradient-echo echo-planar imaging BOLD-fMRI was performed on a 7-Tesla Siemens MRI scanner. Functional images

Sequence & imaging parameters

were acquired using GRAPPA-EPI sequence: echo time = 28 ms, repetition time = 2.34 s, flip angle = 75°, slice orientation = transversal (axial), anterior to posterior phase encoding, voxel size = 1.1 mm isotropic, gap between slices = 0 mm, number of slices = 123, field of view = 205 × 205 mm², GRAPPA acceleration factor = 3; echo spacing = 0.82 ms, bandwidth = 1414 Hz per pixel, partial Fourier in the phase encode direction = 7/8. A custom-built 32-channel radiofrequency coil head array was used for reception. Radiofrequency transmission was provided by a detunable band-pass birdcage coil.

Structural images were acquired using a T1-weighted EPI sequence, selected so that functional and structural images had similar spatial distortions, which facilitated co-registration and subsequent normalization of data to MNI space 38. Structural scan parameters were echo time = 22 ms, repetition time = 8.52 s, flip angle = 90°, number of slices = 126, slice orientation = transversal (axial), voxel size = 1.1 mm isotropic, gap between slices = 0 mm, field of view = 205 × 205 mm², GRAPPA acceleration factor = 3; echo spacing = 0.82 ms, bandwidth = 1414 Hz per pixel, partial Fourier in the phase encode direction = 6/8.

Area of acquisition

Whole brain

Diffusion MRI

☐

Used

☒

Not used

Preprocessing

Preprocessing software

fmripipeline, version 1.1.2; FSL version 5.0.9; ANTs version 2.1.0; AFNI version 16.2.07; SciPy version 1.6

Normalization

Anatomical T1-weighted images were nonlinearly normalized to the ICBM 152 Nonlinear Asymmetrical template (2009c) using brain-extracted images and antsRegistration (ANTs v2.1.0). Functional images were co-registered to the T1w using boundary-based registration, and all transforms (motion correction, BOLD-to-T1w, T1w-to-MNI) were concatenated and applied in a single step using antsApplyTransforms with Lanczos interpolation.

Normalization template

Spatial normalization of anatomical data to the ICBM 152 Nonlinear Asymmetrical template version 2009c

Noise and artifact removal

Motion correction was performed using MCFLIRT, with framewise displacement calculated for each run. GLMs included 24 motion regressors (six translation/rotation parameters, their squares, derivatives, and squared derivatives). Physiological noise was modeled using 10 aCompCor components (five each from CSF and white matter) derived from masks thresholded at 99% probability, with CSF masks constrained to ventricles and white matter masks eroded. Additional regressors included discrete cosine transform terms (minimum 120-second period) and spike regressors censoring volumes with framewise displacement >0.5 mm. Runs were dropped from analysis if > 20% of TRs were excluded via spike regressors.

Volume censoring

Volumes exceeding 0.5 mm framewise displacement were modelled out with spike regressors; runs with more than 20% of TRs flagged were excluded from analysis.

Statistical modeling & inference

Model type and settings

Univariate analysis; mixed effects analysis; multivariate analysis

Effect(s) tested

Univariate Analysis:

First-level GLMs modeled shape presentation, pre-stimulus decision, and sensory stimulation (visual or somatosensory, between-subjects). Contrasts were normalized, smoothed, averaged per subject, and analyzed at the group level using one-sample t-tests (FDR $q < 0.05$ for ROI analyses and $q < 0.01$ for whole-brain analyses).

Mixed Effects Models:

Unsmoothed run-level contrasts within superior colliculus, VPL, and LGN were analyzed using linear mixed effects models including side, voxel depth (SC only), sensory modality, and their interactions, with subject-specific random intercepts and slopes.

Multivariate Pattern Analysis:

Linear SVM classifiers were trained on unsmoothed run-level contrast maps from the superior colliculus using leave-one-subject-out cross-validation across subjects.

Specify type of analysis:

☐

Whole brain

☐

ROI-based

☒

Both

Anatomical location(s)

superior colliculus; lateral geniculate nucleus; ventral posterolateral nucleus; whole-brain

Statistic type for inference

voxel-wise

(See [Eklund et al. 2016](#))

Correction

Group-level univariate analyses applied false discovery rate (FDR) correction at $q < 0.05$ (ROI analyses) or $q < 0.01$ (whole-brain analysis) to control for multiple comparisons.

Models & analysis

n/a	Involvement in the study
☒	☐ Functional and/or effective connectivity
☒	☐ Graph analysis
☐	☒ Multivariate modeling or predictive analysis

Multivariate modeling and predictive analysis

Multivariate modeling employed linear support vector machine (SVM) classifiers trained on voxel-wise activation patterns extracted from unsmoothed run-level contrast maps within the superior colliculus. Models were trained using leave-one-subject-out cross-validation across subjects.